CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

GODINHO DAIRY

12710 S WILSON RD

LOS BANOS CA 93635

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Godinho Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12710 S Wilson RD	Los Banos	Merced	93635

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/09/1990

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X180-X028-XXXX X070-X180-X044-XXXX X070-X180-X045-XXXX

B. OPERATORS

Godinho, David

Operator name: Godinho, David

Telephone no.: (209) 826-2517 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12710 Wilson RD	Los Banos	CA	93635

This operator is responsible for paying permit fees.

C. OWNERS

Godinho, David

Legal owner name: Godinho, David

Telephone no.: (209) 826-2517 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12710 Wilson RD	Los Banos	CA	93635

This owner is responsible for paying permit fees.

Godinho, Manuela

Legal owner name:	Godinho, Manuela	Telephone no.:	(209) 826-2517	
			Landline	Cellular
12710 Wilson RD	Los Banos		CA	93635
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	4,000	700	0	0	0	0
Maximum number	4,000	700	0	0	0	0
Average number	4,000	700	0	0	0	0
Avg live weight (lbs)	1,400	1,500	0	0		

Predominant milk cow breed: Holstein

Average milk production: 84 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 119,586.24 *tons per reporting period*

Total nitrogen from manure: 1,555,200.16 *lbs per reporting period*

After ammonia losses (30% loss applied): 1,088,640.11 *lbs per reporting period*

Total phosphorus from manure: 259,774.46 *lbs per reporting period*

Total potassium from manure: 802,828.91 *lbs per reporting period*

Total salt from manure: 2,044,365.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 22,004,490 *gallons*

Total nitrogen generated: 131,062.82 *lbs*

Total phosphorus generated: 15,552.83 *lbs*

Total potassium generated: 285,090.65 *lbs*

Total salt generated: 1,119,367.86 *lbs*

	33,030,420	*gallons applied*
+	0	*gallons exported*
-	11,025,930	*gallons imported*
=	22,004,490	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID Canal	Surface water
Domestic Well	Ground water
Freshwater Pond	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Corral solids *Solid Manure Imported from Heifer Facility - Spread on Oats 10/2022*	2,092.00 *ton*	Dry-weight	50.9	15,400.00	5,500.00	8,300.00		0.00
05/29/2023	Separator solids *Solid Manure Imported from Heifer Facility - Spread on Corn*	4,316.00 *ton*	Dry-weight	28.8	12,500.00	2,200.00	14,900.00		0.00

Date	Material type / Description	Quantity	N (mg/L)	P (mg/L)	K (mg/L)	Salt (mg/L)	TDS (mg/L)
01/01/2023	Process wastewater *Lagoon Water Imported from Heifer Ranch Q4 2022*	596,520.00 *gal*	587.80	76.10	3,470.00		7,300
01/01/2023	Process wastewater *Lagoon Water Imported From Jose V Silveira Dairy - 11/2022*	4,800,000.00 *gal*	214.00	49.40	200.00		1,950
04/01/2023	Process wastewater *Lagoon Water Imported From Jose V Silveira Dairy*	2,100,000.00 *gal*	596.00	68.20	660.00		4,000
06/04/2023	Process wastewater *Lagoon Water Imported from Heifer Facility Q2*	845,070.00 *gal*	1,281.00	426.00	2,230.00		3,800
08/25/2023	Process wastewater *Lagoon Water Imported from Heifer Facility Q3*	2,684,340.00 *gal*	219.50	79.40	674.00		3,100

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	108,461.70	24,820.06	108,626.22	0.00
Process wastewater	35,893.36	8,335.58	67,675.24	280,786.85
Total imports for all materials	144,355.06	33,155.64	176,301.46	280,786.85

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	44	44	6	both	X070-X180-X028-XXXX
10	77	77	2	both	X070-X200-X010-XXXX
11	40	40	2	manure	X070-X190-X007-XXXX
12N	27	27	2	process wastewater	X070-X180-X021-XXXX
12S	43	43	6	process wastewater	X070-X180-X021-XXXX
13	50	50	2	both	X081-X020-X025-XXXX X081-X020-X026-XXXX X081-X020-X027-XXXX X081-X020-X028-XXXX
14	61	61	2	process wastewater	X070-X180-X008-XXXX X070-X180-X009-XXXX
2	5	5	1	manure	X070-X180-X044-XXXX
5	30	30	2	process wastewater	X070-X180-X045-XXXX
6	30	30	2	process wastewater	X070-X180-X050-XXXX
7	156	156	2	manure	X078-X130-X015-XXXX
8	18	18	2	manure	X078-X130-X015-XXXX
9	108	108	2	manure	X078-X130-X060-XXXX
E1	38	38	2	both	X070-X201-X011-XXXX
Fahey 1	113	97	2	manure	X070-X100-X008-XXXX
Fahey 2	175	175	2	manure	X070-X100-X029-XXXX X070-X100-X030-XXXX
Fahey 3	167	112	2	both	X070-X090-X035-XXXX X070-X090-X036-XXXX
JM1	80	80	2	manure	X070-X160-X055-XXXX
JM2	80	80	2	manure	X070-X160-X055-XXXX
JM3	60	60	2	manure	X070-X160-X032-XXXX X070-X160-X034-XXXX X070-X160-X039-XXXX X070-X160-X040-XXXX X070-X160-X041-XXXX
JM4	65	65	2	manure	X070-X160-X032-XXXX X070-X160-X034-XXXX X070-X160-X039-XXXX X070-X160-X040-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
					X070-X160-X041-XXXX
JM5	65	65	2	manure	X070-X180-X002-XXXX
Totals for areas that were used for application	1,532	1,461	51		
Totals for areas that were not used for application					
Land application area totals	1,532	1,461	51		

B. CROPS AND HARVESTS

1

Field name: 1

09/20/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 09/20/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	70.40 *ton*	Dry-weight		7.2	29,050.00	2,800.00	25,050.00		11.40
05/25/2023	25.60 *ton*	Dry-weight		6.1	35,000.00	2,300.00	18,900.00		8.90
06/22/2023	17.00 *ton*	Dry-weight		6.1	35,000.00	2,300.00	18,900.00		8.90
07/20/2023	18.30 *ton*	Dry-weight		7.2	29,050.00	2,800.00	25,050.00		11.40
08/17/2023	19.20 *ton*	Dry-weight		8.2	23,100.00	3,300.00	31,200.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	3.76	209.92	19.03	168.40	769.69

08/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 40 Plant date: 08/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	398.80 *ton*	Dry-weight		75.4	17,100.00	1,700.00	32,100.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	9.97	83.88	8.34	157.46	480.71

10

Field name: 10

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	897.20 *ton*	Dry-weight		60.4	20,600.00	2,950.00	28,400.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.65	190.10	27.22	262.08	913.61

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,906.00 *ton*	Dry-weight		54.4	14,900.00	1,550.00	26,700.00		7.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.75	336.37	34.99	602.75	1,726.98

11

Field name: 11

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	465.60 *ton*	Dry-weight		65.4	10,400.00	1,900.00	14,850.00		10.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.64	83.77	15.30	119.61	809.52

11

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,023.00 *ton*	Dry-weight		65.4	9,400.00	2,700.00	11,100.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.58	166.36	47.78	196.45	955.69

12N

Field name: 12N

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	291.70 *ton*	Dry-weight		63.6	19,700.00	2,800.00	25,500.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.80	154.94	22.02	200.56	637.07

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	660.00 *ton*	Dry-weight		60.0	16,100.00	1,700.00	20,500.00		7.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.44	314.84	33.24	400.89	1,544.89

12S

Field name: 12S

09/20/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 09/20/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	70.40 *ton*	Dry-weight		6.2	31,600.00	2,800.00	22,000.00		10.50
05/25/2023	38.30 *ton*	Dry-weight		5.9	32,000.00	2,300.00	15,500.00		8.40
06/22/2023	27.40 *ton*	Dry-weight		5.9	32,000.00	2,300.00	15,500.00		8.40
07/20/2023	28.10 *ton*	Dry-weight		6.2	31,600.00	2,800.00	22,000.00		10.50
08/17/2023	26.80 *ton*	Dry-weight		6.5	31,200.00	3,300.00	28,500.00		12.60
09/19/2023	27.60 *ton*	Dry-weight		6.5	31,200.00	3,300.00	28,500.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.47	324.25	28.44	222.03	1,065.17

13

Field name: 13

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	526.30 *ton*	Dry-weight		64.6	15,500.00	2,600.00	21,000.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.16	144.39	24.22	195.63	968.81

13

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	812.00 *ton*	Dry-weight		53.6	12,300.00	2,500.00	14,800.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.30	231.71	47.10	278.81	1,130.30

14

Field name: 14

11/21/2022: Ryegrass, hay

Crop: Ryegrass, hay Acres planted: 61 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/07/2023	310.00 *ton*	Dry-weight		45.4	17,550.00	2,500.00	29,750.00		10.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	192.00	27.60	252.00	0.00
Total actual harvest content	5.08	97.39	13.87	165.10	591.02

07/27/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 61 Plant date: 07/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	503.60 *ton*	Dry-weight		67.5	15,800.00	2,000.00	32,300.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.26	84.79	10.73	173.33	660.05

2

Field name: 2

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 5 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	50.00 *ton*	Dry-weight		59.6	17,200.00	3,700.00	30,800.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	138.98	29.90	248.86	921.12

5

Field name: 5

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	381.50 *ton*	Dry-weight		56.4	14,650.00	2,150.00	21,200.00		7.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.72	162.45	23.84	235.09	859.39

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	734.00 *ton*	Dry-weight		56.1	15,400.00	2,100.00	20,850.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.47	330.82	45.11	447.89	1,503.72

6

Field name: 6

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	381.50 *ton*	Dry-weight		56.4	14,650.00	2,150.00	21,200.00		7.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.72	162.45	23.84	235.09	859.39

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	734.00 *ton*	Dry-weight		56.1	15,400.00	2,100.00	20,850.00		7.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.47	330.82	45.11	447.89	1,503.72

7

Field name: 7

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 156 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,953.00 *ton*	Dry-weight		55.7	8,100.00	2,150.00	12,700.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.52	89.85	23.85	140.87	887.36

7

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 156 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	2,551.10 *ton*	Dry-weight		73.9	13,650.00	3,950.00	29,850.00		13.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	16.35	116.52	33.72	254.81	1,182.29

8

Field name: 8

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	220.00 *ton*	Dry-weight		60.4	8,600.00	2,000.00	13,000.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.22	83.25	19.36	125.84	919.60

06/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 18 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	305.10 *ton*	Dry-weight		73.5	16,400.00	4,000.00	29,800.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	16.95	147.33	35.93	267.71	1,167.86

9

Field name: 9

10/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 108 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,344.30 *ton*	Dry-weight		50.2	6,700.00	1,700.00	12,700.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.45	83.06	21.08	157.45	843.03

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 108 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	2,176.00 *ton*	Dry-weight		55.0	10,350.00	2,650.00	9,150.00		5.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.15	187.68	48.05	165.92	933.87

E1

Field name: E1

11/24/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	534.60 *ton*	Dry-weight		68.1	24,150.00	2,700.00	26,800.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.07	216.76	24.23	240.55	789.86

E1

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	792.00 *ton*	Dry-weight		62.7	11,100.00	1,700.00	10,400.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.84	172.59	26.43	161.70	653.02

Fahey 1

Field name: Fahey 1

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 97 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,485.50 *ton*	Dry-weight		60.0	8,150.00	1,900.00	12,800.00		8.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.31	99.85	23.28	156.82	1,072.01

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 97 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	2,521.00 *ton*	Dry-weight		54.8	10,600.00	3,500.00	7,500.00		3.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.99	249.04	82.23	176.21	869.30

Fahey 2

Field name: Fahey 2

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 175 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	2,695.40 *ton*	Dry-weight		73.4	16,533.00	3,467.00	28,067.00		10.43

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.40	135.47	28.41	229.98	854.64

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 175 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	4,374.00 *ton*	Dry-weight		52.5	9,467.00	3,033.00	10,900.00		5.03

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.99	224.79	72.02	258.82	1,194.35

Fahey 3

Field name: Fahey 3

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 112 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,710.20 *ton*	Dry-weight		67.1	15,233.00	2,967.00	25,233.00		9.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.27	153.05	29.81	253.53	937.42

Fahey 3

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 112 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	2,898.00 *ton*	Dry-weight		49.6	11,700.00	2,800.00	10,650.00		4.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.88	305.16	73.03	277.77	1,238.90

JM1

Field name: JM1

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 73 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,086.90 *ton*	Dry-weight		69.9	12,350.00	2,300.00	19,050.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.89	110.70	20.62	170.75	752.91

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 73 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,668.00 *ton*	Dry-weight		60.4	11,750.00	2,250.00	7,850.00		3.95

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.85	212.64	40.72	142.06	714.82

JM2

Field name: JM2

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 72 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	939.30 *ton*	Dry-weight		61.2	9,200.00	1,800.00	12,850.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.05	93.14	18.22	130.09	931.37

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,628.00 *ton*	Dry-weight		66.8	11,950.00	2,150.00	9,000.00		5.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.61	179.41	32.28	135.12	758.20

JM3

Field name: JM3

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	597.10 *ton*	Dry-weight		66.9	24,050.00	4,550.00	31,350.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.86	172.85	32.70	225.31	848.06

JM3

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,244.70 *ton*	Dry-weight		58.6	11,350.00	2,100.00	7,600.00		3.95

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.63	212.68	39.35	142.41	740.17

JM4

Field name: JM4

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 59 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	640.50 *ton*	Dry-weight		66.9	24,050.00	4,550.00	31,350.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.86	172.84	32.70	225.30	848.02

05/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 05/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,335.30 *ton*	Dry-weight		58.6	11,350.00	2,100.00	7,600.00		3.95

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.63	212.69	39.35	142.42	740.21

JM5

Field name: JM5

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 59 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	626.70 *ton*	Dry-weight		62.9	10,800.00	2,150.00	13,800.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.62	85.12	16.95	108.77	669.93

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,376.00 *ton*	Dry-weight		65.4	10,550.00	1,600.00	8,300.00		4.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.32	170.26	25.82	133.95	702.04

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 09/20/2019: Alfalfa, hay

Field name: 1

Crop: Alfalfa, hay

Plant date: 09/20/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.63	0.00	0.00	179.26	4,296,338.00 *gal*
Application event totals		0.63	0.00	0.00	179.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.66	17.44	130.46	912.63	994,200.00 *gal*
CCID Canal	Surface water	1.51	0.00	0.00	430.72	10,322,777.00 *gal*
Application event totals		115.17	17.44	130.46	1,343.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	132.57	13.10	164.07	969.87	695,940.00 *gal*
CCID Canal	Surface water	1.73	0.00	0.00	492.95	11,814,327.00 *gal*
Application event totals		134.30	13.10	164.07	1,462.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.84	0.00	0.00	241.00	5,775,840.00 *gal*
Application event totals		0.84	0.00	0.00	241.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.21	0.00	0.00	346.46	8,303,522.00 *gal*
Application event totals		1.21	0.00	0.00	346.46	

1 - 08/20/2023: Sudangrass, silage

Field name: 1

Crop: Sudangrass, silage — Plant date: 08/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	112.29	17.70	77.89	1,861.34	360.00 *ton*
Application event totals		112.29	17.70	77.89	1,861.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.86	0.00	0.00	245.42	5,881,862.00 *gal*
Application event totals		0.86	0.00	0.00	245.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.26	6,501,133.00 *gal*
Application event totals		0.95	0.00	0.00	271.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.39	0.00	0.00	112.10	2,686,717.00 *gal*
Application event totals		0.39	0.00	0.00	112.10	

10 - 10/29/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

10 - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Corral solids	67.76	24.20	36.52	0.00	345.00 *ton*
Application event totals		67.76	24.20	36.52	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.33	13.12	598.22	1,258.50	1,590,720.00 *gal*
CCID Canal	Surface water	0.73	0.00	0.00	208.72	9,629,532.00 *gal*
Application event totals		102.07	13.12	598.22	1,467.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.57	13.59	101.66	711.14	1,491,300.00 *gal*
CCID Canal	Surface water	0.63	0.00	0.00	179.26	8,270,450.00 *gal*
Application event totals		89.20	13.59	101.66	890.40	

10 - 06/01/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Separator solids	132.92	23.39	158.44	0.00	575.00 *ton*
Application event totals		132.92	23.39	158.44	0.00	

10 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.51	0.00	0.00	430.72	19,871,346.00 *gal*
Application event totals		1.51	0.00	0.00	430.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.30	10.20	127.84	755.74	1,043,910.00 *gal*
CCID Canal	Surface water	1.73	0.00	0.00	492.95	22,742,579.00 *gal*
Application event totals		105.03	10.20	127.84	1,248.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.84	0.00	0.00	241.00	11,118,491.00 *gal*
Application event totals		0.84	0.00	0.00	241.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.21	0.00	0.00	346.46	15,984,281.00 *gal*
Application event totals		1.21	0.00	0.00	346.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.22	10.69	133.93	791.73	1,093,620.00 *gal*
CCID Canal	Surface water	0.86	0.00	0.00	245.42	11,322,585.00 *gal*
Application event totals		109.08	10.69	133.93	1,037.15	

10 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.22	10.69	133.93	791.73	1,093,620.00 *gal*
CCID Canal	Surface water	0.95	0.00	0.00	271.26	12,514,680.00 *gal*
Application event totals		109.17	10.69	133.93	1,062.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	3.69	0.00	0.00	299.56	9,698,400.00 *gal*
CCID Canal	Surface water	0.39	0.00	0.00	112.10	5,171,931.00 *gal*
Application event totals		4.08	0.00	0.00	411.66	

11 - 11/01/2022: Oats, silage-soft dough

Field name: 11

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	107.95	13.96	46.53	0.00	391.00 *ton*
Application event totals		107.95	13.96	46.53	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.78	0.00	0.00	222.28	5,327,144.00 *gal*
Application event totals		0.78	0.00	0.00	222.28	

11 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.59	0.00	0.00	169.94	4,072,869.00 *gal*
Application event totals		0.59	0.00	0.00	169.94	

11 - 05/20/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	217.40	34.27	150.81	3,603.77	697.00 *ton*
Application event totals		217.40	34.27	150.81	3,603.77	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.63	0.00	0.00	465.13	11,147,542.00 *gal*
Application event totals		1.63	0.00	0.00	465.13	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.40	0.00	0.00	400.74	9,604,361.00 *gal*
Application event totals		1.40	0.00	0.00	400.74	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.41	0.00	0.00	403.39	9,667,779.00 *gal*
Application event totals		1.41	0.00	0.00	403.39	

11 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.38	0.00	0.00	395.45	9,477,524.00 *gal*
Application event totals		1.38	0.00	0.00	395.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.48	0.00	0.00	422.21	10,118,755.00 *gal*
Application event totals		1.48	0.00	0.00	422.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	317.54	7,610,205.00 *gal*
Application event totals		1.11	0.00	0.00	317.54	

12N - 11/20/2022: Oats, silage-soft dough

Field name: 12N

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.31	11.69	533.13	1,121.58	497,100.00 *gal*
Irrigation Well	Ground water	12.07	0.00	0.00	1,893.99	3,829,970.00 *gal*
Application event totals		102.38	11.69	533.13	3,015.57	

12N - 11/20/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.61	14.21	106.30	743.62	546,810.00 *gal*
Irrigation Well	Ground water	10.70	0.00	0.00	1,678.76	3,394,746.00 *gal*
CCID Canal	Surface water	0.11	0.00	0.00	30.85	499,065.00 *gal*
Application event totals		103.42	14.21	106.30	2,453.24	

12N - 05/21/2023: Corn, silage

Field name: 12N

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.03	15.51	115.97	811.22	596,520.00 *gal*
Irrigation Well	Ground water	17.01	0.00	0.00	2,668.81	5,396,776.00 *gal*
Application event totals		118.05	15.51	115.97	3,480.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.90	0.00	0.00	2,023.13	4,091,104.00 *gal*
CCID Canal	Surface water	0.09	0.00	0.00	25.17	407,132.00 *gal*
Application event totals		12.99	0.00	0.00	2,048.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.23	11.09	138.89	821.06	397,680.00 *gal*
Irrigation Well	Ground water	12.62	0.00	0.00	1,980.08	4,004,060.00 *gal*
CCID Canal	Surface water	0.15	0.00	0.00	43.03	696,064.00 *gal*
Application event totals		125.01	11.09	138.89	2,844.16	

12N - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.72	0.00	0.00	2,152.26	4,352,239.00 *gal*
CCID Canal	Surface water	0.06	0.00	0.00	17.05	275,799.00 *gal*
Application event totals		13.78	0.00	0.00	2,169.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	105.22	10.39	130.21	769.74	372,825.00 *gal*
Irrigation Well	Ground water	14.82	0.00	0.00	2,324.44	4,700,418.00 *gal*
CCID Canal	Surface water	0.23	0.00	0.00	66.57	1,076,929.00 *gal*
Application event totals		120.27	10.39	130.21	3,160.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.00	0.00	0.00	2,195.31	4,439,284.00 *gal*
CCID Canal	Surface water	0.22	0.00	0.00	63.32	1,024,396.00 *gal*
Application event totals		14.22	0.00	0.00	2,258.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.90	0.00	0.00	2,023.13	4,091,104.00 *gal*
CCID Canal	Surface water	0.08	0.00	0.00	22.73	367,732.00 *gal*
Application event totals		12.98	0.00	0.00	2,045.86	

12S - 09/20/2019: Alfalfa, hay

Field name: 12S

Crop: Alfalfa, hay

Plant date: 09/20/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

12S - 09/20/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.70	0.00	0.00	1,678.76	5,029,254.00 *gal*
CCID Canal	Surface water	0.11	0.00	0.00	30.85	739,355.00 *gal*
Application event totals		10.81	0.00	0.00	1,709.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.66	17.44	130.46	912.63	994,200.00 *gal*
Irrigation Well	Ground water	17.01	0.00	0.00	2,668.81	7,995,224.00 *gal*
Application event totals		130.68	17.44	130.46	3,581.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	132.57	13.10	164.07	969.87	695,940.00 *gal*
Irrigation Well	Ground water	12.90	0.00	0.00	2,023.13	6,060,896.00 *gal*
CCID Canal	Surface water	0.09	0.00	0.00	25.17	603,158.00 *gal*
Application event totals		145.56	13.10	164.07	3,018.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.62	0.00	0.00	1,980.08	5,931,940.00 *gal*
CCID Canal	Surface water	0.15	0.00	0.00	43.03	1,031,206.00 *gal*
Application event totals		12.77	0.00	0.00	2,023.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.72	0.00	0.00	2,152.26	6,447,761.00 *gal*
CCID Canal	Surface water	0.06	0.00	0.00	17.05	408,591.00 *gal*
Application event totals		13.78	0.00	0.00	2,169.31	

12S - 09/20/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.82	0.00	0.00	2,324.44	6,963,582.00 *gal*
CCID Canal	Surface water	0.23	0.00	0.00	66.57	1,595,451.00 *gal*
Application event totals		15.05	0.00	0.00	2,391.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.00	0.00	0.00	2,195.31	6,576,716.00 *gal*
CCID Canal	Surface water	0.22	0.00	0.00	63.32	1,517,624.00 *gal*
Application event totals		14.22	0.00	0.00	2,258.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.90	0.00	0.00	2,023.13	6,060,896.00 *gal*
CCID Canal	Surface water	0.08	0.00	0.00	22.73	544,788.00 *gal*
Application event totals		12.98	0.00	0.00	2,045.86	

13 - 10/28/2022: Oats, silage-soft dough

Field name: 13

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	73.15	9.47	431.84	908.48	596,520.00 *gal*
CCID Canal	Surface water	0.73	0.00	0.00	208.05	4,986,270.00 *gal*
Application event totals		73.88	9.47	431.84	1,116.53	

13 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	112.92	37.55	196.58	334.98	422,535.00 *gal*
CCID Canal	Surface water	0.73	0.00	0.00	208.05	4,986,270.00 *gal*
Application event totals		113.65	37.55	196.58	543.03	

13 - 06/01/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Separator solids	68.09	11.98	81.16	0.00	153.00 *ton*
Application event totals		68.09	11.98	81.16	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	112.92	37.55	196.58	334.98	422,535.00 *gal*
CCID Canal	Surface water	1.14	0.00	0.00	325.00	7,789,010.00 *gal*
Application event totals		114.06	37.55	196.58	659.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	36.42	13.17	111.84	514.39	795,360.00 *gal*
CCID Canal	Surface water	1.28	0.00	0.00	366.47	8,783,005.00 *gal*
Application event totals		37.70	13.17	111.84	880.86	

13 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.34	0.00	0.00	382.79	9,174,085.00 *gal*
Application event totals		1.34	0.00	0.00	382.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.79	0.00	0.00	227.09	5,442,530.00 *gal*
Application event totals		0.79	0.00	0.00	227.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	45.53	16.47	139.80	642.99	994,200.00 *gal*
Freshwater Pond	Surface water	3.31	0.00	0.00	269.10	4,525,920.00 *gal*
CCID Canal	Surface water	0.40	0.00	0.00	114.22	2,737,560.00 *gal*
Application event totals		49.24	16.47	139.80	1,026.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water - Imported from Heifer Facility	Process wastewater	40.97	14.82	125.82	578.69	894,780.00 *gal*
Freshwater Pond	Surface water	3.39	0.00	0.00	275.51	4,633,680.00 *gal*
CCID Canal	Surface water	0.30	0.00	0.00	85.67	2,053,170.00 *gal*
Application event totals		44.67	14.82	125.82	939.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.63	0.00	0.00	179.50	4,301,880.00 *gal*
Application event totals		0.63	0.00	0.00	179.50	

13 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.68	0.00	0.00	193.09	4,627,780.00 *gal*
Application event totals		0.68	0.00	0.00	193.09	

14 - 11/21/2022: Ryegrass, hay

Field name: 14

Crop: Ryegrass, hay

Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.57	12.24	558.26	1,174.43	1,176,000.00 *gal*
Application event totals		94.57	12.24	558.26	1,174.43	

11/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.85	0.00	0.00	1,702.05	7,776,000.00 *gal*
Application event totals		10.85	0.00	0.00	1,702.05	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.04	0.00	0.00	1,418.38	6,480,000.00 *gal*
CCID Canal	Surface water	0.24	0.00	0.00	69.55	2,542,020.00 *gal*
Application event totals		9.29	0.00	0.00	1,487.93	

14 - 07/27/2023: Sudangrass, silage

Field name: 14

Crop: Sudangrass, silage

Plant date: 07/27/2023

14 - 07/27/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.66	0.00	0.00	1,985.73	9,072,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	133.75	4,888,500.00 *gal*
Application event totals		13.13	0.00	0.00	2,119.48	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.26	0.00	0.00	2,080.29	9,504,000.00 *gal*
Application event totals		13.26	0.00	0.00	2,080.29	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	12.66	0.00	0.00	1,985.73	9,072,000.00 *gal*
CCID Canal	Surface water	0.41	0.00	0.00	115.92	4,236,700.00 *gal*
Application event totals		13.06	0.00	0.00	2,101.65	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.47	0.00	0.00	2,269.40	10,368,000.00 *gal*
Application event totals		14.47	0.00	0.00	2,269.40	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.77	0.00	0.00	2,316.68	10,584,000.00 *gal*
CCID Canal	Surface water	0.07	0.00	0.00	21.40	782,160.00 *gal*
Application event totals		14.84	0.00	0.00	2,338.08	

14 - 07/27/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	14.47	0.00	0.00	2,269.40	10,368,000.00 *gal*
Application event totals		14.47	0.00	0.00	2,269.40	

10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.07	0.00	0.00	2,363.96	10,800,000.00 *gal*
Application event totals		15.07	0.00	0.00	2,363.96	

2 - 11/01/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Corral solids	166.35	59.41	89.66	0.00	55.00 *ton*
Application event totals		166.35	59.41	89.66	0.00	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.98	0.00	0.00	565.68	1,694,680.00 *gal*
Application event totals		1.98	0.00	0.00	565.68	

5 - 10/29/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 10/29/2022

5 - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.79	14.73	671.75	1,413.19	695,940.00 *gal*
Freshwater Pond	Surface water	3.05	0.00	0.00	247.75	3,125,040.00 *gal*
CCID Canal	Surface water	0.43	0.00	0.00	123.29	2,216,120.00 *gal*
Application event totals		117.27	14.73	671.75	1,784.22	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.93	13.95	104.37	730.10	596,520.00 *gal*
Freshwater Pond	Surface water	3.16	0.00	0.00	256.29	3,232,800.00 *gal*
Application event totals		94.09	13.95	104.37	986.39	

5 - 05/31/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	3.79	0.00	0.00	307.55	3,879,360.00 *gal*
CCID Canal	Surface water	0.33	0.00	0.00	94.28	1,694,680.00 *gal*
Application event totals		4.12	0.00	0.00	401.83	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	101.01	9.98	125.00	738.95	397,680.00 *gal*
Freshwater Pond	Surface water	4.63	0.00	0.00	375.89	4,741,440.00 *gal*
CCID Canal	Surface water	0.14	0.00	0.00	40.79	733,275.00 *gal*
Application event totals		105.78	9.98	125.00	1,155.63	

5 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.32	10.60	132.81	785.13	422,535.00 *gal*
Freshwater Pond	Surface water	4.84	0.00	0.00	392.98	4,956,960.00 *gal*
CCID Canal	Surface water	0.16	0.00	0.00	44.42	798,455.00 *gal*
Application event totals		112.32	10.60	132.81	1,222.53	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.32	10.60	132.81	785.13	422,535.00 *gal*
Freshwater Pond	Surface water	4.52	0.00	0.00	367.35	4,633,680.00 *gal*
CCID Canal	Surface water	0.23	0.00	0.00	65.27	1,173,240.00 *gal*
Application event totals		112.07	10.60	132.81	1,217.75	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.38	8.73	109.38	646.58	347,970.00 *gal*
Freshwater Pond	Surface water	4.84	0.00	0.00	392.98	4,956,960.00 *gal*
CCID Canal	Surface water	0.11	0.00	0.00	32.64	586,620.00 *gal*
Application event totals		93.34	8.73	109.38	1,072.19	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	4.73	0.00	0.00	384.43	4,849,200.00 *gal*
CCID Canal	Surface water	0.09	0.00	0.00	24.48	439,965.00 *gal*
Application event totals		4.82	0.00	0.00	408.91	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	5.16	0.00	0.00	418.60	5,280,240.00 *gal*
Application event totals		5.16	0.00	0.00	418.60	

6 - 10/29/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.79	14.73	671.75	1,413.19	695,940.00 *gal*
Freshwater Pond	Surface water	3.05	0.00	0.00	247.75	3,125,040.00 *gal*
CCID Canal	Surface water	0.43	0.00	0.00	123.29	2,216,120.00 *gal*
Application event totals		117.27	14.73	671.75	1,784.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.93	13.95	104.37	730.10	596,520.00 *gal*
Freshwater Pond	Surface water	3.16	0.00	0.00	256.29	3,232,800.00 *gal*
Application event totals		94.09	13.95	104.37	986.39	

6 - 05/31/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	3.79	0.00	0.00	307.55	3,879,360.00 *gal*
CCID Canal	Surface water	0.33	0.00	0.00	94.28	1,694,680.00 *gal*
Application event totals		4.12	0.00	0.00	401.83	

6 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.01	9.98	125.00	738.95	397,680.00 *gal*
Freshwater Pond	Surface water	4.63	0.00	0.00	375.89	4,741,440.00 *gal*
CCID Canal	Surface water	0.14	0.00	0.00	40.79	733,275.00 *gal*
Application event totals		105.78	9.98	125.00	1,155.63	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.32	10.60	132.81	785.13	422,535.00 *gal*
Freshwater Pond	Surface water	4.84	0.00	0.00	392.98	4,956,960.00 *gal*
CCID Canal	Surface water	0.16	0.00	0.00	44.42	798,455.00 *gal*
Application event totals		112.32	10.60	132.81	1,222.53	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.32	10.60	132.81	785.13	422,535.00 *gal*
Freshwater Pond	Surface water	4.52	0.00	0.00	367.35	4,633,680.00 *gal*
CCID Canal	Surface water	0.23	0.00	0.00	65.27	1,173,240.00 *gal*
Application event totals		112.07	10.60	132.81	1,217.75	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.38	8.73	109.38	646.58	347,970.00 *gal*
Freshwater Pond	Surface water	4.84	0.00	0.00	392.98	4,956,960.00 *gal*
CCID Canal	Surface water	0.11	0.00	0.00	32.64	586,620.00 *gal*
Application event totals		93.34	8.73	109.38	1,072.19	

6 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	4.73	0.00	0.00	384.43	4,849,200.00 *gal*
CCID Canal	Surface water	0.09	0.00	0.00	24.48	439,965.00 *gal*
Application event totals		4.82	0.00	0.00	408.91	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	5.16	0.00	0.00	418.60	5,280,240.00 *gal*
Application event totals		5.16	0.00	0.00	418.60	

7 - 11/10/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	117.16	15.15	50.50	0.00	1,655.00 *ton*
Application event totals		117.16	15.15	50.50	0.00	

11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.48	0.00	0.00	137.91	12,890,385.00 *gal*
Application event totals		0.48	0.00	0.00	137.91	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	316.71	29,602,814.00 *gal*
Application event totals		1.11	0.00	0.00	316.71	

7 - 06/15/2023: Sudangrass, silage

Field name: 7

Crop: Sudangrass, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	147.96	23.33	102.64	2,452.63	1,850.00 *ton*
Application event totals		147.96	23.33	102.64	2,452.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.66	0.00	0.00	474.68	44,368,165.00 *gal*
Application event totals		1.66	0.00	0.00	474.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.62	0.00	0.00	463.11	43,286,454.00 *gal*
Application event totals		1.62	0.00	0.00	463.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.44	0.00	0.00	410.84	38,400,728.00 *gal*
Application event totals		1.44	0.00	0.00	410.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.35	0.00	0.00	385.76	36,057,021.00 *gal*
Application event totals		1.35	0.00	0.00	385.76	

7 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.85	0.00	0.00	529.08	49,452,205.00 *gal*
Application event totals		1.85	0.00	0.00	529.08	

8 - 10/24/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	107.36	13.88	46.28	0.00	175.00 *ton*
Application event totals		107.36	13.88	46.28	0.00	

11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.48	0.00	0.00	137.91	1,487,352.00 *gal*
Application event totals		0.48	0.00	0.00	137.91	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	316.71	3,415,709.00 *gal*
Application event totals		1.11	0.00	0.00	316.71	

8 - 06/15/2023: Sudangrass, silage

Field name: 8

Crop: Sudangrass, silage

Plant date: 06/15/2023

8 - 06/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	187.15	29.51	129.82	3,102.24	270.00 *ton*
Application event totals		187.15	29.51	129.82	3,102.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.66	0.00	0.00	474.68	5,119,404.00 *gal*
Application event totals		1.66	0.00	0.00	474.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.62	0.00	0.00	463.11	4,994,591.00 *gal*
Application event totals		1.62	0.00	0.00	463.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.44	0.00	0.00	410.84	4,430,853.00 *gal*
Application event totals		1.44	0.00	0.00	410.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.35	0.00	0.00	385.76	4,160,426.00 *gal*
Application event totals		1.35	0.00	0.00	385.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.85	0.00	0.00	529.08	5,706,024.00 *gal*
Application event totals		1.85	0.00	0.00	529.08	

9 - 10/24/2022: Oats, silage-soft dough

9 - 10/24/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	107.67	13.92	46.41	0.00	1,053.00 *ton*
Application event totals		107.67	13.92	46.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.48	0.00	0.00	137.91	8,924,113.00 *gal*
Application event totals		0.48	0.00	0.00	137.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	316.71	20,494,256.00 *gal*
Application event totals		1.11	0.00	0.00	316.71	

9 - 05/28/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	242.60	38.25	168.29	4,021.42	2,100.00 *ton*
Application event totals		242.60	38.25	168.29	4,021.42	

9 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.66	0.00	0.00	474.68	30,716,422.00 *gal*
Application event totals		1.66	0.00	0.00	474.68	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.62	0.00	0.00	463.11	29,967,545.00 *gal*
Application event totals		1.62	0.00	0.00	463.11	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.44	0.00	0.00	410.84	26,585,119.00 *gal*
Application event totals		1.44	0.00	0.00	410.84	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.35	0.00	0.00	385.76	24,962,553.00 *gal*
Application event totals		1.35	0.00	0.00	385.76	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.85	0.00	0.00	529.08	34,236,142.00 *gal*
Application event totals		1.85	0.00	0.00	529.08	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.31	0.00	0.00	373.70	24,181,780.00 *gal*
Application event totals		1.31	0.00	0.00	373.70	

E1 - 11/24/2022: Oats, silage-soft dough

E1 - 11/24/2022: Oats, silage-soft dough

Field name: E1

Crop: Oats, silage-soft dough

Plant date: 11/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Corral solids	87.55	31.27	47.19	0.00	220.00 *ton*
Application event totals		87.55	31.27	47.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.08	14.12	643.97	1,354.75	845,070.00 *gal*
CCID Canal	Surface water	0.77	0.00	0.00	219.71	5,002,354.00 *gal*
Application event totals		109.85	14.12	643.97	1,574.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.71	15.15	113.30	792.54	820,215.00 *gal*
CCID Canal	Surface water	0.66	0.00	0.00	188.70	4,296,338.00 *gal*
Application event totals		99.37	15.15	113.30	981.24	

E1 - 06/02/2023: Corn, silage

Field name: E1

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.82	9.18	68.67	480.33	497,100.00 *gal*
CCID Canal	Surface water	1.59	0.00	0.00	453.39	10,322,777.00 *gal*
Application event totals		61.41	9.18	68.67	933.72	

E1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.74	7.88	98.69	583.38	397,680.00 *gal*
CCID Canal	Surface water	1.82	0.00	0.00	518.90	11,814,327.00 *gal*
Application event totals		81.56	7.88	98.69	1,102.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.74	7.88	98.69	583.38	397,680.00 *gal*
CCID Canal	Surface water	0.89	0.00	0.00	253.68	5,775,840.00 *gal*
Application event totals		80.63	7.88	98.69	837.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.28	0.00	0.00	364.70	8,303,522.00 *gal*
Application event totals		1.28	0.00	0.00	364.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.90	0.00	0.00	258.34	5,881,862.00 *gal*
Application event totals		0.90	0.00	0.00	258.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.54	6,501,133.00 *gal*
Application event totals		1.00	0.00	0.00	285.54	

E1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Freshwater Pond	Surface water	3.74	0.00	0.00	303.50	4,849,200.00 *gal*
CCID Canal	Surface water	0.41	0.00	0.00	118.00	2,686,717.00 *gal*
Application event totals		4.15	0.00	0.00	421.50	

Fahey 1 - 10/25/2022: Oats, silage-soft dough

Field name: Fahey 1

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Corral solids	65.79	23.50	35.46	0.00	422.00 *ton*
Solid Manure	Compost	65.69	8.49	28.31	0.00	577.00 *ton*
Application event totals		131.48	31.99	63.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.73	0.00	0.00	208.72	12,130,709.00 *gal*
Application event totals		0.73	0.00	0.00	208.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.63	0.00	0.00	179.26	10,418,619.00 *gal*
Application event totals		0.63	0.00	0.00	179.26	

Fahey 1 - 05/30/2023: Corn, silage

Fahey 1 - 05/30/2023: Corn, silage

Field name: Fahey 1

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Separator solids	164.60	28.97	196.21	0.00	897.00 *ton*
Solid Manure	Corral solids	164.64	25.96	114.21	2,729.12	1,280.00 *ton*
Application event totals		329.24	54.93	310.42	2,729.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.51	0.00	0.00	430.72	25,032,734.00 *gal*
Application event totals		1.51	0.00	0.00	430.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.73	0.00	0.00	492.95	28,649,742.00 *gal*
Application event totals		1.73	0.00	0.00	492.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.84	0.00	0.00	241.00	14,006,411.00 *gal*
Application event totals		0.84	0.00	0.00	241.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.21	0.00	0.00	346.46	20,136,042.00 *gal*
Application event totals		1.21	0.00	0.00	346.46	

Fahey 1 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.86	0.00	0.00	245.42	14,263,517.00 *gal*
Application event totals		0.86	0.00	0.00	245.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.26	15,765,247.00 *gal*
Application event totals		0.95	0.00	0.00	271.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.92	0.00	0.00	646.73	11,220,000.00 *gal*
CCID Canal	Surface water	0.39	0.00	0.00	112.10	6,515,289.00 *gal*
Application event totals		5.32	0.00	0.00	758.83	

Fahey 2 - 10/26/2022: Oats, silage-soft dough

Field name: Fahey 2

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Corral solids	90.74	32.41	48.90	0.00	1,050.00 *ton*
Solid Manure	Compost	90.68	11.73	39.09	0.00	1,437.00 *ton*
Application event totals		181.42	44.13	87.99	0.00	

Fahey 2 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.73	0.00	0.00	208.72	21,885,299.00 *gal*
Application event totals		0.73	0.00	0.00	208.72	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.63	0.00	0.00	179.26	18,796,478.00 *gal*
Application event totals		0.63	0.00	0.00	179.26	

Fahey 2 - 06/03/2023: Corn, silage

Field name: Fahey 2

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Separator solids	145.45	25.60	173.38	0.00	1,430.00 *ton*
Solid Manure	Corral solids	145.80	22.99	101.14	2,416.79	2,045.00 *ton*
Application event totals		291.25	48.59	274.52	2,416.79	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.51	0.00	0.00	430.72	45,162,150.00 *gal*
Application event totals		1.51	0.00	0.00	430.72	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.73	0.00	0.00	492.95	51,687,680.00 *gal*
Application event totals		1.73	0.00	0.00	492.95	

Fahey 2 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.84	0.00	0.00	241.00	25,269,298.00 *gal*
Application event totals		0.84	0.00	0.00	241.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.21	0.00	0.00	346.46	36,327,910.00 *gal*
Application event totals		1.21	0.00	0.00	346.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.86	0.00	0.00	245.42	25,733,148.00 *gal*
Application event totals		0.86	0.00	0.00	245.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.26	28,442,456.00 *gal*
Application event totals		0.95	0.00	0.00	271.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.55	0.00	0.00	684.53	16,500,000.00 *gal*
CCID Canal	Surface water	0.39	0.00	0.00	112.10	11,754,388.00 *gal*
Application event totals		7.95	0.00	0.00	796.63	

Fahey 3 - 10/26/2022: Oats, silage-soft dough

Field name: Fahey 3

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Fahey 3 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Imported	Process wastewater	76.86	17.67	71.53	697.40	4,800,000.00 *gal*
CCID Canal	Surface water	0.73	0.00	0.00	208.72	14,006,592.00 *gal*
Application event totals		77.59	17.67	71.53	906.13	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water Imported	Process wastewater	93.46	10.67	103.27	625.88	2,100,000.00 *gal*
CCID Canal	Surface water	0.63	0.00	0.00	179.26	12,029,746.00 *gal*
Application event totals		94.09	10.67	103.27	805.14	

Fahey 3 - 06/03/2023: Corn, silage

Field name: Fahey 3

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Heifer Ranch	Separator solids	200.25	35.24	238.70	0.00	1,260.00 *ton*
Solid Manure	Corral solids	189.37	29.86	131.37	3,139.17	1,700.00 *ton*
Application event totals		389.62	65.10	370.07	3,139.17	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.51	0.00	0.00	430.72	28,903,776.00 *gal*
Application event totals		1.51	0.00	0.00	430.72	

Fahey 3 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.73	0.00	0.00	492.95	33,080,115.00 *gal*
Application event totals		1.73	0.00	0.00	492.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.84	0.00	0.00	241.00	16,172,351.00 *gal*
Application event totals		0.84	0.00	0.00	241.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.21	0.00	0.00	346.46	23,249,863.00 *gal*
Application event totals		1.21	0.00	0.00	346.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.86	0.00	0.00	245.42	16,469,215.00 *gal*
Application event totals		0.86	0.00	0.00	245.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.26	18,203,172.00 *gal*
Application event totals		0.95	0.00	0.00	271.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.58	0.00	0.00	777.87	12,000,000.00 *gal*
CCID Canal	Surface water	0.39	0.00	0.00	112.10	7,522,808.00 *gal*
Application event totals		8.98	0.00	0.00	889.98	

JM1 - 11/22/2022: Oats, silage-soft dough

Field name: JM1

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	146.74	18.97	63.25	0.00	970.00 *ton*
Application event totals		146.74	18.97	63.25	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.78	0.00	0.00	222.28	9,722,037.00 *gal*
Application event totals		0.78	0.00	0.00	222.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.59	0.00	0.00	169.94	7,432,986.00 *gal*
Application event totals		0.59	0.00	0.00	169.94	

JM1 - 05/22/2023: Corn, silage

Field name: JM1

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	245.26	38.67	170.13	4,065.49	1,435.00 *ton*
Application event totals		245.26	38.67	170.13	4,065.49	

JM1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.63	0.00	0.00	465.13	20,344,263.00 *gal*
Application event totals		1.63	0.00	0.00	465.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.40	0.00	0.00	400.74	17,527,959.00 *gal*
Application event totals		1.40	0.00	0.00	400.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.41	0.00	0.00	403.39	17,643,698.00 *gal*
Application event totals		1.41	0.00	0.00	403.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.38	0.00	0.00	395.45	17,296,482.00 *gal*
Application event totals		1.38	0.00	0.00	395.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.48	0.00	0.00	422.21	18,466,727.00 *gal*
Application event totals		1.48	0.00	0.00	422.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	317.54	13,888,625.00 *gal*
Application event totals		1.11	0.00	0.00	317.54	

JM2 - 11/22/2022: Oats, silage-soft dough

JM2 - 11/22/2022: Oats, silage-soft dough

Field name: JM2

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	121.17	15.67	52.23	0.00	790.00 *ton*
Application event totals		121.17	15.67	52.23	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.78	0.00	0.00	222.28	9,588,859.00 *gal*
Application event totals		0.78	0.00	0.00	222.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.59	0.00	0.00	169.94	7,331,165.00 *gal*
Application event totals		0.59	0.00	0.00	169.94	

JM2 - 05/22/2023: Corn, silage

Field name: JM2

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	235.67	37.15	163.48	3,906.52	1,360.00 *ton*
Application event totals		235.67	37.15	163.48	3,906.52	

JM2 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.63	0.00	0.00	465.13	20,065,575.00 *gal*
Application event totals		1.63	0.00	0.00	465.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.40	0.00	0.00	400.74	17,287,850.00 *gal*
Application event totals		1.40	0.00	0.00	400.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.41	0.00	0.00	403.39	17,402,003.00 *gal*
Application event totals		1.41	0.00	0.00	403.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.38	0.00	0.00	395.45	17,059,544.00 *gal*
Application event totals		1.38	0.00	0.00	395.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.48	0.00	0.00	422.21	18,213,758.00 *gal*
Application event totals		1.48	0.00	0.00	422.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.11	0.00	0.00	317.54	13,698,370.00 *gal*
Application event totals		1.11	0.00	0.00	317.54	

JM3 - 11/22/2022: Oats, silage-soft dough

JM3 - 11/22/2022: Oats, silage-soft dough

Field name: JM3

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	219.86	28.43	94.77	0.00	1,095.00 *ton*
Application event totals		219.86	28.43	94.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.01	8,930,804.00 *gal*
Application event totals		0.95	0.00	0.00	271.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.35	0.00	0.00	791.14	7,344,000.00 *gal*
Application event totals		7.35	0.00	0.00	791.14	

JM3 - 05/21/2023: Corn, silage

Field name: JM3

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.35	0.00	0.00	791.14	7,344,000.00 *gal*
CCID Canal	Surface water	0.28	0.00	0.00	79.68	2,625,782.00 *gal*
Application event totals		7.63	0.00	0.00	870.82	

JM3 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	223.44	35.23	155.00	3,703.89	985.00 *ton*
Application event totals		223.44	35.23	155.00	3,703.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.65	0.00	0.00	930.76	8,640,000.00 *gal*
Application event totals		8.65	0.00	0.00	930.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.14	0.00	0.00	767.87	7,128,000.00 *gal*
Application event totals		7.14	0.00	0.00	767.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.14	0.00	0.00	767.87	7,128,000.00 *gal*
CCID Canal	Surface water	0.10	0.00	0.00	27.67	911,948.00 *gal*
Application event totals		7.23	0.00	0.00	795.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.79	0.00	0.00	837.68	7,776,000.00 *gal*
CCID Canal	Surface water	0.17	0.00	0.00	47.24	1,556,601.00 *gal*
Application event totals		7.95	0.00	0.00	884.92	

JM3 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.71	0.00	0.00	721.34	6,696,000.00 *gal*
CCID Canal	Surface water	0.16	0.00	0.00	46.76	1,540,878.00 *gal*
Application event totals		6.87	0.00	0.00	768.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.49	0.00	0.00	698.07	6,480,000.00 *gal*
CCID Canal	Surface water	0.22	0.00	0.00	62.50	2,059,745.00 *gal*
Application event totals		6.71	0.00	0.00	760.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.79	0.00	0.00	837.68	7,776,000.00 *gal*
CCID Canal	Surface water	0.24	0.00	0.00	68.71	2,264,147.00 *gal*
Application event totals		8.03	0.00	0.00	906.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.14	0.00	0.00	767.87	7,128,000.00 *gal*
CCID Canal	Surface water	0.08	0.00	0.00	23.86	786,162.00 *gal*
Application event totals		7.22	0.00	0.00	791.73	

JM4 - 11/22/2022: Oats, silage-soft dough

Field name: JM4

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

JM4 - 11/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	216.18	27.95	93.18	0.00	1,155.00 *ton*
Application event totals		216.18	27.95	93.18	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.95	0.00	0.00	271.01	9,580,316.00 *gal*
Application event totals		0.95	0.00	0.00	271.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.06	0.00	0.00	759.20	7,560,000.00 *gal*
Application event totals		7.06	0.00	0.00	759.20	

JM4 - 05/21/2023: Corn, silage

Field name: JM4

Crop: Corn, silage

Plant date: 05/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.26	0.00	0.00	780.89	7,776,000.00 *gal*
CCID Canal	Surface water	0.28	0.00	0.00	79.68	2,816,748.00 *gal*
Application event totals		7.54	0.00	0.00	860.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	223.09	35.17	154.76	3,698.15	1,055.00 *ton*
Application event totals		223.09	35.17	154.76	3,698.15	

JM4 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.47	0.00	0.00	911.04	9,072,000.00 *gal*
Application event totals		8.47	0.00	0.00	911.04	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.06	0.00	0.00	759.20	7,560,000.00 *gal*
Application event totals		7.06	0.00	0.00	759.20	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.06	0.00	0.00	759.20	7,560,000.00 *gal*
CCID Canal	Surface water	0.10	0.00	0.00	27.67	978,272.00 *gal*
Application event totals		7.15	0.00	0.00	786.87	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.66	0.00	0.00	824.27	8,208,000.00 *gal*
CCID Canal	Surface water	0.17	0.00	0.00	47.24	1,669,809.00 *gal*
Application event totals		7.83	0.00	0.00	871.51	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.86	0.00	0.00	737.51	7,344,000.00 *gal*
CCID Canal	Surface water	0.16	0.00	0.00	46.76	1,652,942.00 *gal*
Application event totals		7.02	0.00	0.00	784.26	

JM4 - 05/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.65	0.00	0.00	715.81	7,128,000.00 *gal*
CCID Canal	Surface water	0.22	0.00	0.00	62.50	2,209,545.00 *gal*
Application event totals		6.87	0.00	0.00	778.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.86	0.00	0.00	845.96	8,424,000.00 *gal*
CCID Canal	Surface water	0.24	0.00	0.00	68.71	2,428,813.00 *gal*
Application event totals		8.10	0.00	0.00	914.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.26	0.00	0.00	780.89	7,776,000.00 *gal*
CCID Canal	Surface water	0.08	0.00	0.00	23.86	843,338.00 *gal*
Application event totals		7.34	0.00	0.00	804.74	

JM5 - 11/23/2022: Oats, silage-soft dough

Field name: JM5

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	110.43	14.28	47.60	0.00	590.00 *ton*
Application event totals		110.43	14.28	47.60	0.00	

JM5 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.91	0.00	0.00	259.98	9,190,380.00 *gal*
Application event totals		0.91	0.00	0.00	259.98	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.62	0.00	0.00	177.01	6,257,280.00 *gal*
Application event totals		0.62	0.00	0.00	177.01	

JM5 - 05/24/2023: Corn, silage

Field name: JM5

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	219.92	34.67	152.56	3,645.57	1,040.00 *ton*
Application event totals		219.92	34.67	152.56	3,645.57	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.93	0.00	0.00	265.97	9,402,215.00 *gal*
Application event totals		0.93	0.00	0.00	265.97	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.58	0.00	0.00	451.74	15,969,100.00 *gal*
Application event totals		1.58	0.00	0.00	451.74	

JM5 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.48	0.00	0.00	424.08	14,991,400.00 *gal*
Application event totals		1.48	0.00	0.00	424.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.45	0.00	0.00	413.02	14,600,320.00 *gal*
Application event totals		1.45	0.00	0.00	413.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.45	0.00	0.00	413.02	14,600,320.00 *gal*
Application event totals		1.45	0.00	0.00	413.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.68	0.00	0.00	479.39	16,946,800.00 *gal*
Application event totals		1.68	0.00	0.00	479.39	

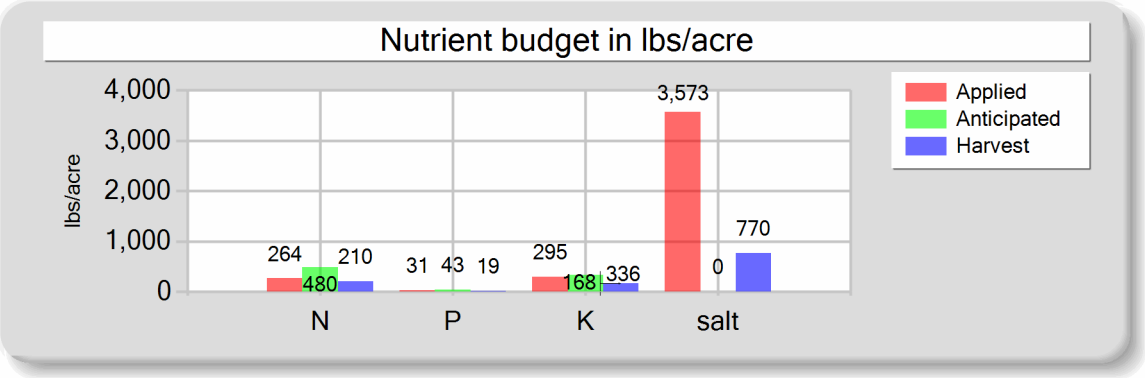
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.93	0.00	0.00	265.97	9,402,215.00 *gal*
Application event totals		0.93	0.00	0.00	265.97	

B. NUTRIENT BUDGET

1 - 09/20/2019: Alfalfa, hay

Field name: 1 Crop: Alfalfa, hay Plant date: 09/20/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.24	30.54	294.53	1,882.50
Fresh water	5.92	0.00	0.00	1,690.40
Atmospheric deposition	11.67	0.00	0.00	0.00
Total nutrients applied	263.82	30.54	294.53	3,572.90
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	209.92	19.03	168.40	769.69
Nutrient balance	53.90	11.51	126.13	2,803.20
Applied to removed ratio	1.26	1.60	1.75	4.64

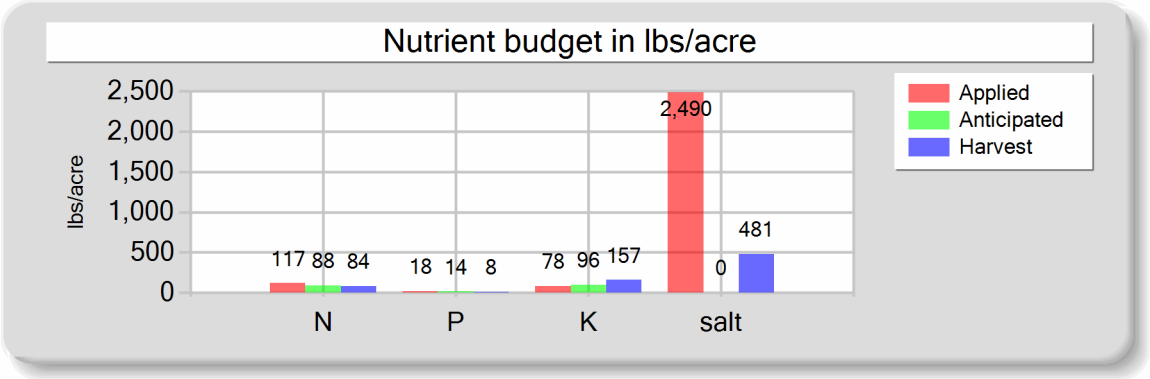
Fresh water applied	
40,512,804.00	*gallons*
1,491.95	*acre-inches*
37.30	*inches/acre*

Process wastewater applied	
1,690,140.00	*gallons*
62.24	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
5	*harvests*

1 - 08/20/2023: Sudangrass, silage

Field name: 1 Crop: Sudangrass, silage Plant date: 08/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	112.29	17.70	77.89	1,861.34
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.20	0.00	0.00	628.78
Atmospheric deposition	2.33	0.00	0.00	0.00
Total nutrients applied	116.82	17.70	77.89	2,490.13
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	83.88	8.34	157.46	480.71
Nutrient balance	32.94	9.36	-79.57	2,009.41
Applied to removed ratio	1.39	2.12	0.49	5.18

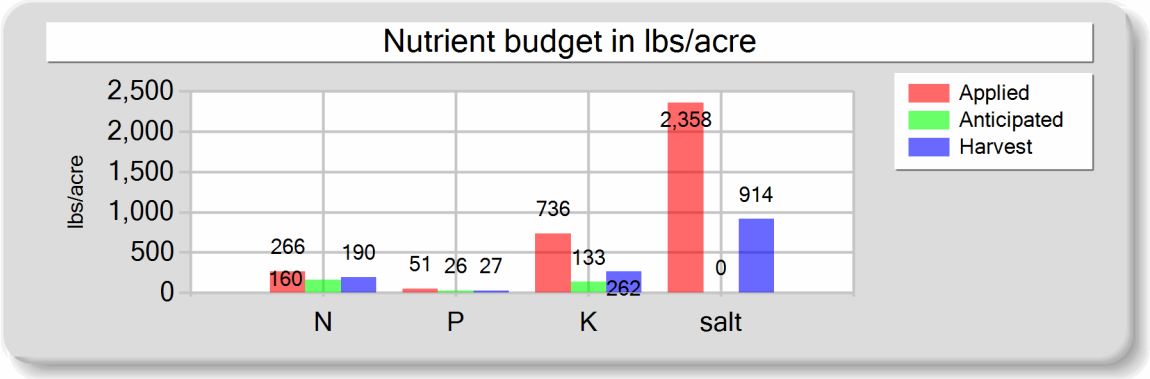
Fresh water applied	
15,069,712.00	*gallons*
554.97	*acre-inches*
13.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 10/29/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	67.76	24.20	36.52	0.00
Process wastewater	189.90	26.71	699.88	1,969.63
Fresh water	1.36	0.00	0.00	387.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.02	50.91	736.40	2,357.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	190.10	27.22	262.08	913.61
Nutrient balance	75.92	23.69	474.31	1,444.02
Applied to removed ratio	1.40	1.87	2.81	2.58

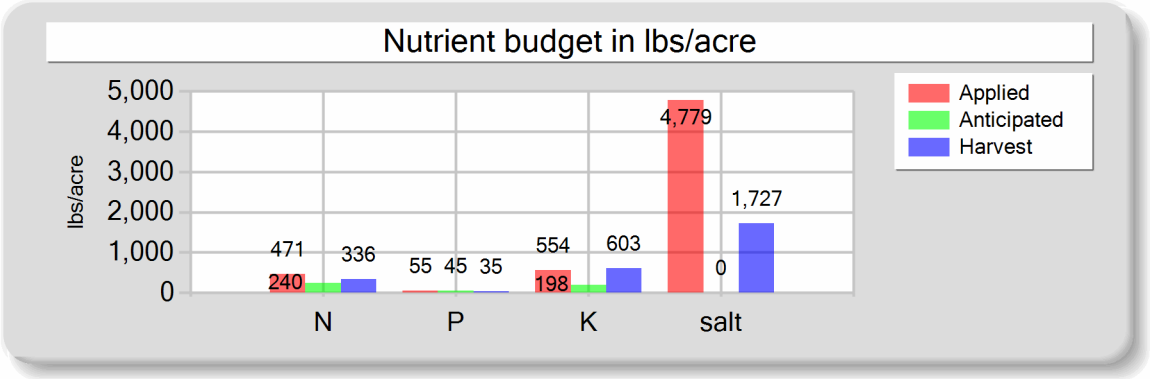
Fresh water applied	
17,899,982.00	*gallons*
659.20	*acre-inches*
8.56	*inches/acre*

Process wastewater applied	
3,082,020.00	*gallons*
113.50	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/01/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	132.92	23.39	158.44	0.00
Process wastewater	319.75	31.59	395.70	2,339.21
Fresh water	11.18	0.00	0.00	2,439.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	470.85	54.98	554.15	4,778.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	336.37	34.99	602.75	1,726.98
Nutrient balance	134.48	19.99	-48.60	3,051.70
Applied to removed ratio	1.40	1.57	0.92	2.77

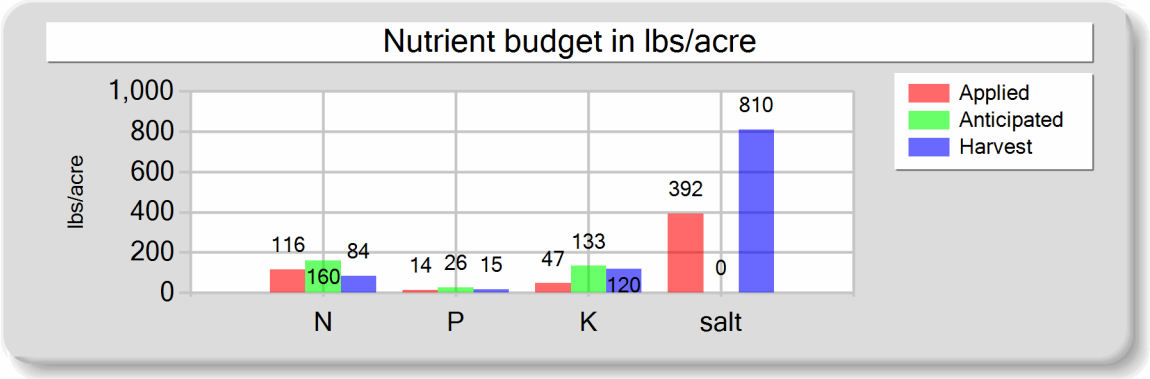
Fresh water applied	
108,424,293.00	*gallons*
3,992.90	*acre-inches*
51.86	*inches/acre*

Process wastewater applied	
3,231,150.00	*gallons*
118.99	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/01/2022: Oats, silage-soft dough

Field name: 11 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.95	13.96	46.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.37	0.00	0.00	392.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.32	13.96	46.53	392.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	83.77	15.30	119.61	809.52
Nutrient balance	32.55	-1.35	-73.09	-417.30
Applied to removed ratio	1.39	0.91	0.39	0.48

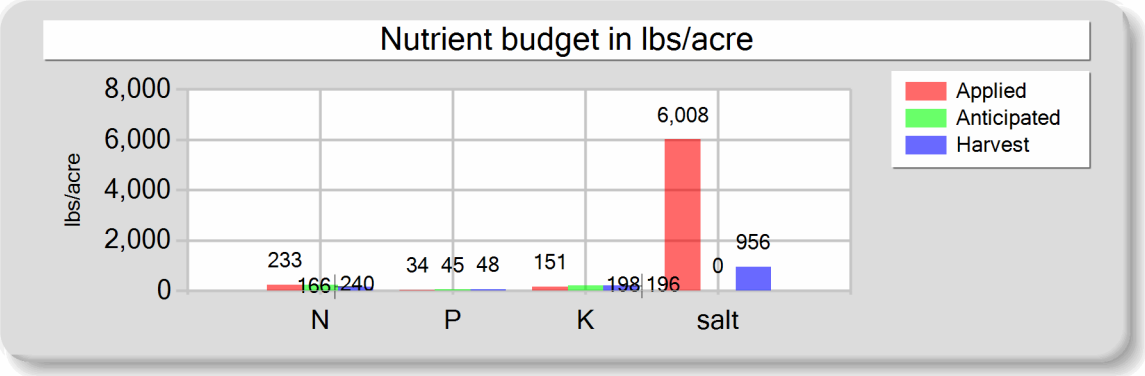
Fresh water applied	
9,400,013.00	*gallons*
346.17	*acre-inches*
8.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 05/20/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	217.40	34.27	150.81	3,603.77
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	2,404.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.82	34.27	150.81	6,008.22
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	166.36	47.78	196.45	955.69
Nutrient balance	66.46	-13.51	-45.64	5,052.53
Applied to removed ratio	1.40	0.72	0.77	6.29

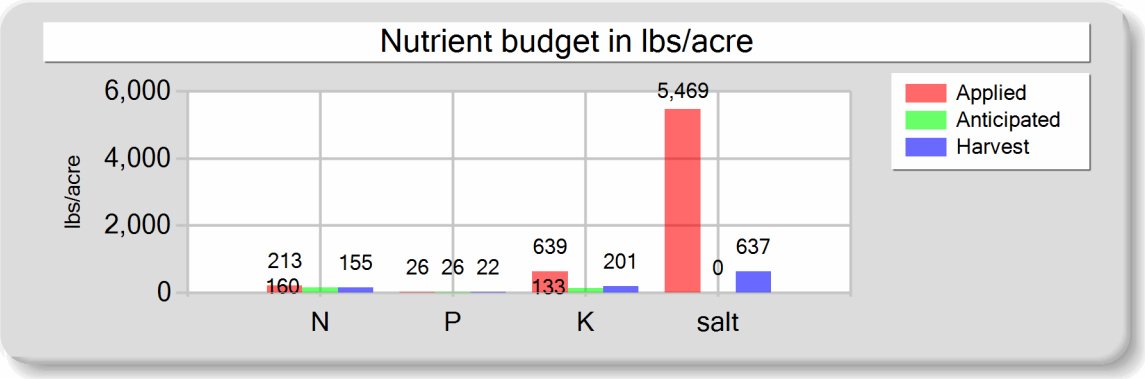
Fresh water applied	
57,626,166.00	*gallons*
2,122.18	*acre-inches*
53.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12N - 11/20/2022: Oats, silage-soft dough

Field name: 12N | Crop: Oats, silage-soft dough | Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.92	25.91	639.44	1,865.20
Fresh water	22.88	0.00	0.00	3,603.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.81	25.91	639.44	5,468.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.94	22.02	200.56	637.07
Nutrient balance	57.87	3.88	438.88	4,831.73
Applied to removed ratio	1.37	1.18	3.19	8.58

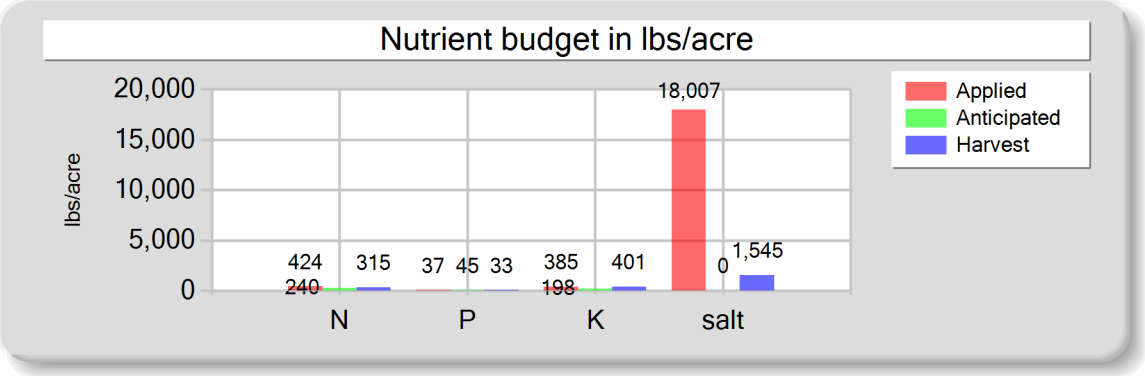
Fresh water applied	
7,723,781.00	*gallons*
284.44	*acre-inches*
10.53	*inches/acre*

Process wastewater applied	
1,043,910.00	*gallons*
38.44	*acre-inches*
1.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

12N - 05/21/2023: Corn, silage

Field name: 12N Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	318.48	36.99	385.07	2,402.02
Fresh water	98.80	0.00	0.00	15,605.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	424.28	36.99	385.07	18,007.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	314.84	33.24	400.89	1,544.89
Nutrient balance	109.44	3.74	-15.82	16,462.15
Applied to removed ratio	1.35	1.11	0.96	11.66

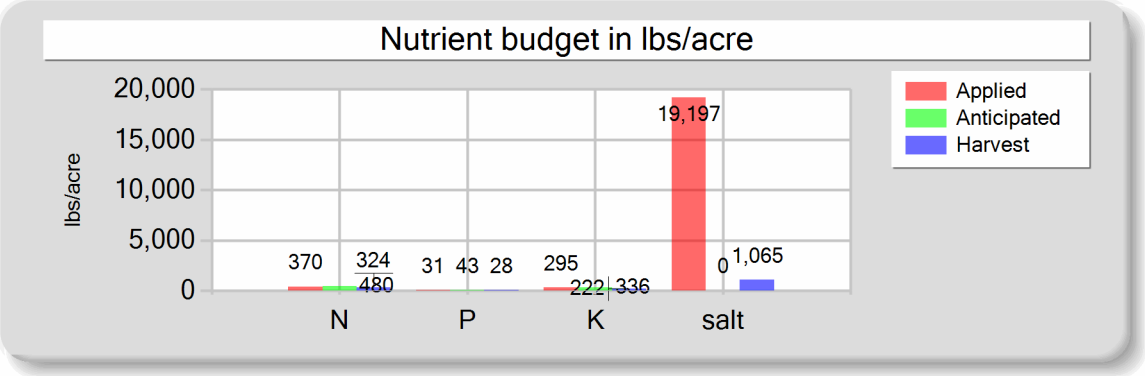
Fresh water applied	
34,923,037.00	*gallons*
1,286.10	*acre-inches*
47.63	*inches/acre*

Process wastewater applied	
1,367,025.00	*gallons*
50.34	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

12S - 09/20/2019: Alfalfa, hay

Field name: 12S | Crop: Alfalfa, hay | Plant date: 09/20/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	246.24	30.54	294.53	1,882.50
Fresh water	109.61	0.00	0.00	17,314.64
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	369.85	30.54	294.53	19,197.14
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	324.25	28.44	222.03	1,065.17
Nutrient balance	45.60	2.10	72.50	18,131.97
Applied to removed ratio	1.14	1.07	1.33	18.02

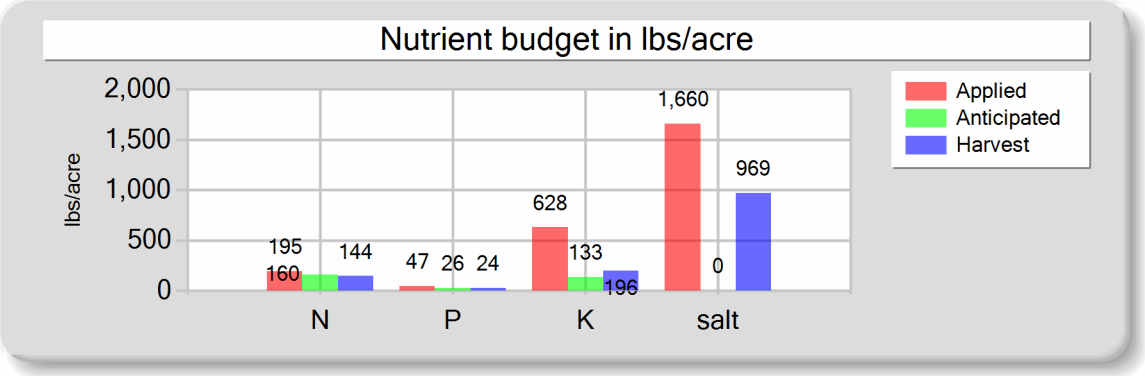
Fresh water applied	
57,506,442.00	*gallons*
2,117.77	*acre-inches*
52.94	*inches/acre*

Process wastewater applied	
1,690,140.00	*gallons*
62.24	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
6	*harvests*

13 - 10/28/2022: Oats, silage-soft dough

Field name: 13 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.07	47.02	628.42	1,243.45
Fresh water	1.46	0.00	0.00	416.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.53	47.02	628.42	1,659.56
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	144.39	24.22	195.63	968.81
Nutrient balance	50.14	22.80	432.79	690.74
Applied to removed ratio	1.35	1.94	3.21	1.71

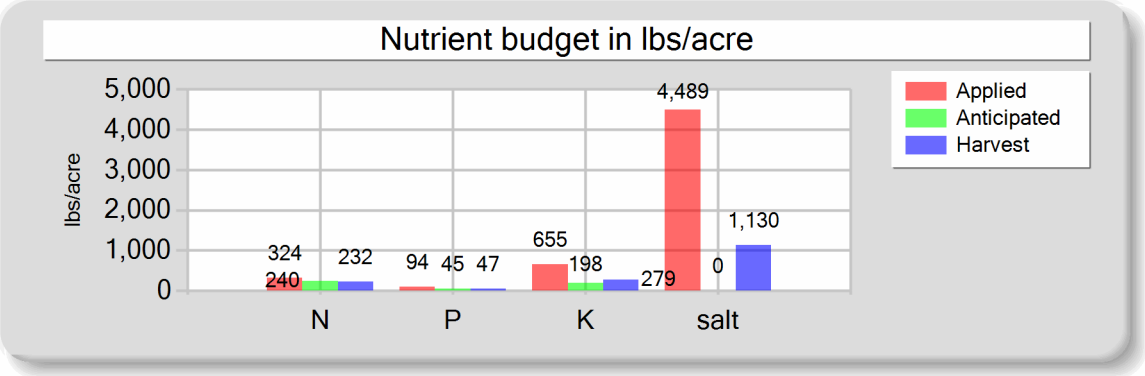
Fresh water applied	
9,972,540.00	*gallons*
367.25	*acre-inches*
9.18	*inches/acre*

Process wastewater applied	
1,019,055.00	*gallons*
37.53	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 06/01/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	68.09	11.98	81.16	0.00
Process wastewater	235.85	82.02	574.03	2,071.04
Fresh water	13.27	0.00	0.00	2,418.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.20	94.00	655.19	4,489.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.71	47.10	278.81	1,130.30
Nutrient balance	92.48	46.91	376.38	3,359.18
Applied to removed ratio	1.40	2.00	2.35	3.97

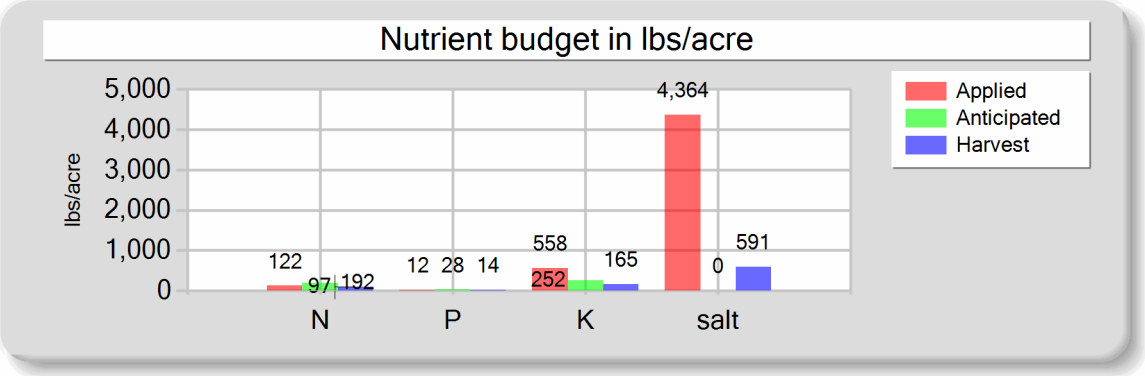
Fresh water applied	
54,068,620.00	*gallons*
1,991.16	*acre-inches*
49.78	*inches/acre*

Process wastewater applied	
3,106,875.00	*gallons*
114.42	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 11/21/2022: Ryegrass, hay

Field name: 14 Crop: Ryegrass, hay Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.57	12.24	558.26	1,174.43
Fresh water	20.14	0.00	0.00	3,189.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	121.70	12.24	558.26	4,364.41
Anticipated crop nutrient removal	192.00	27.60	252.00	0.00
Actual crop nutrient removal	97.39	13.87	165.10	591.02
Nutrient balance	24.31	-1.63	393.16	3,773.39
Applied to removed ratio	1.25	0.88	3.38	7.38

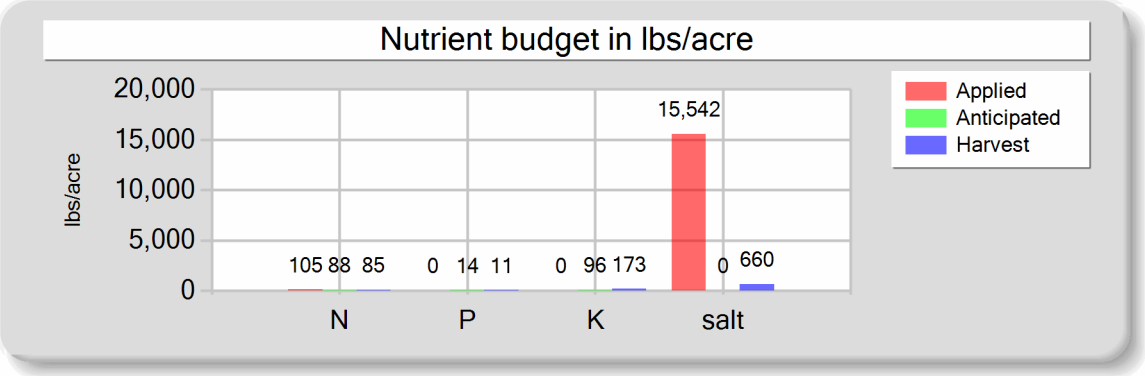
Fresh water applied	
16,798,020.00	*gallons*
618.61	*acre-inches*
10.14	*inches/acre*

Process wastewater applied	
1,176,000.00	*gallons*
43.31	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 07/27/2023: Sudangrass, silage

Field name: 14 Crop: Sudangrass, silage Plant date: 07/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	98.30	0.00	0.00	15,542.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	105.30	0.00	0.00	15,542.26
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	84.79	10.73	173.33	660.05
Nutrient balance	20.52	-10.73	-173.33	14,882.21
Applied to removed ratio	1.24	0.00	0.00	23.55

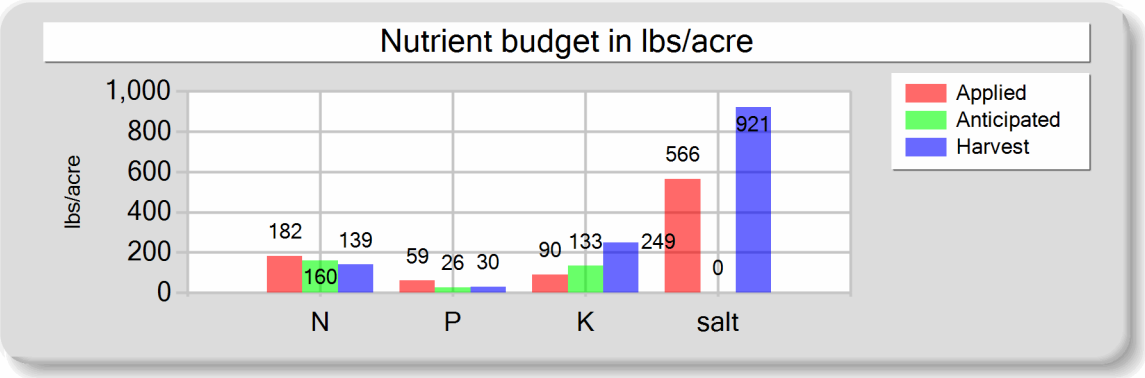
Fresh water applied	
79,675,360.00	*gallons*
2,934.17	*acre-inches*
48.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/01/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.35	59.41	89.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.98	0.00	0.00	565.68
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	182.33	59.41	89.66	565.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.98	29.90	248.86	921.12
Nutrient balance	43.35	29.52	-159.21	-355.44
Applied to removed ratio	1.31	1.99	0.36	0.61

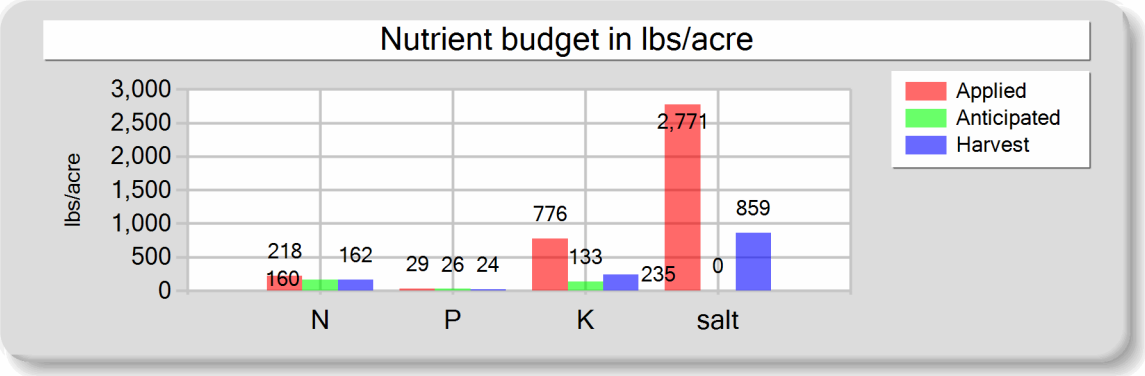
Fresh water applied	
1,694,680.00	*gallons*
62.41	*acre-inches*
12.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/29/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.72	28.69	776.12	2,143.29
Fresh water	6.64	0.00	0.00	627.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.36	28.69	776.12	2,770.61
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.45	23.84	235.09	859.39
Nutrient balance	55.91	4.85	541.03	1,911.22
Applied to removed ratio	1.34	1.20	3.30	3.22

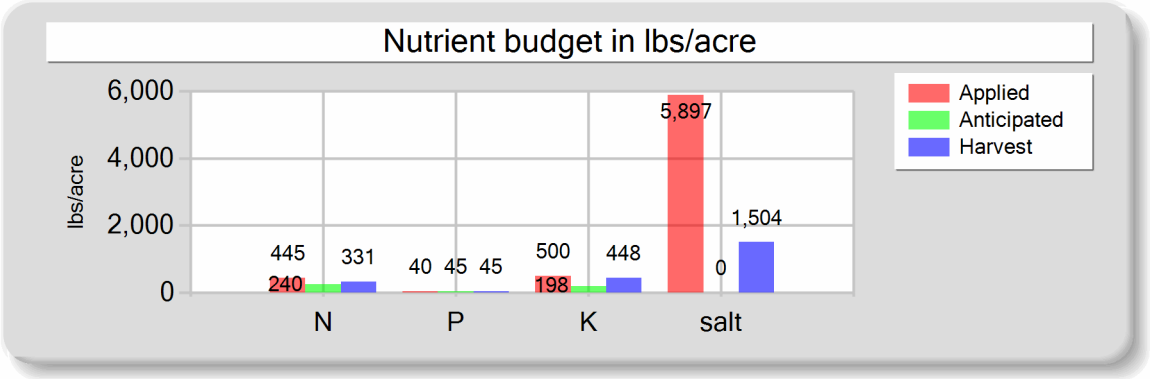
Fresh water applied	
8,573,960.00	*gallons*
315.75	*acre-inches*
10.52	*inches/acre*

Process wastewater applied	
1,292,460.00	*gallons*
47.60	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/31/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	404.03	39.91	500.01	2,955.80
Fresh water	33.57	0.00	0.00	2,941.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	444.60	39.91	500.01	5,897.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	330.82	45.11	447.89	1,503.72
Nutrient balance	113.78	-5.20	52.11	4,393.73
Applied to removed ratio	1.34	0.88	1.12	3.92

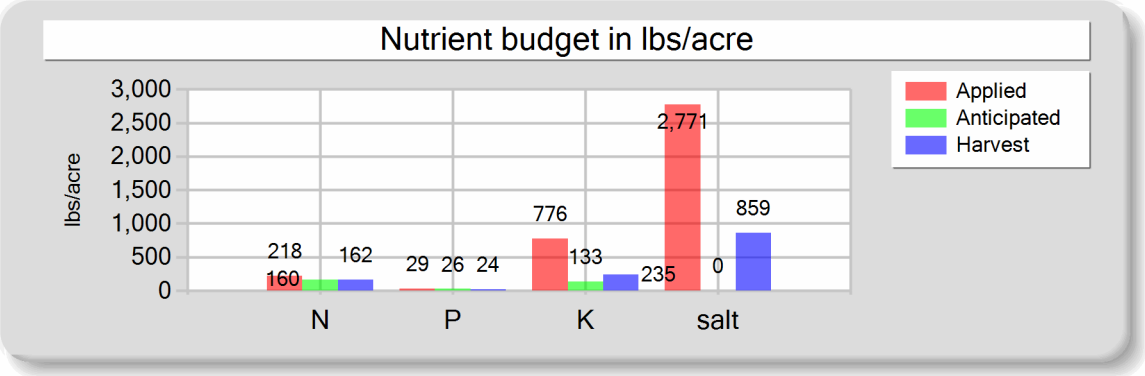
Fresh water applied	
38,724,075.00	*gallons*
1,426.08	*acre-inches*
47.54	*inches/acre*

Process wastewater applied	
1,590,720.00	*gallons*
58.58	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 10/29/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.72	28.69	776.12	2,143.29
Fresh water	6.64	0.00	0.00	627.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.36	28.69	776.12	2,770.61
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.45	23.84	235.09	859.39
Nutrient balance	55.91	4.85	541.03	1,911.22
Applied to removed ratio	1.34	1.20	3.30	3.22

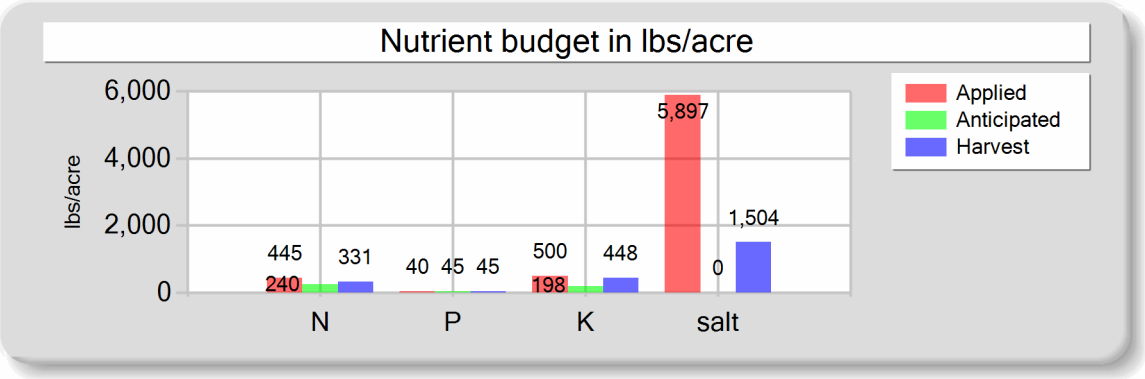
Fresh water applied	
8,573,960.00	*gallons*
315.75	*acre-inches*
10.52	*inches/acre*

Process wastewater applied	
1,292,460.00	*gallons*
47.60	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 05/31/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	404.03	39.91	500.01	2,955.80
Fresh water	33.57	0.00	0.00	2,941.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	444.60	39.91	500.01	5,897.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	330.82	45.11	447.89	1,503.72
Nutrient balance	113.78	-5.20	52.11	4,393.73
Applied to removed ratio	1.34	0.88	1.12	3.92

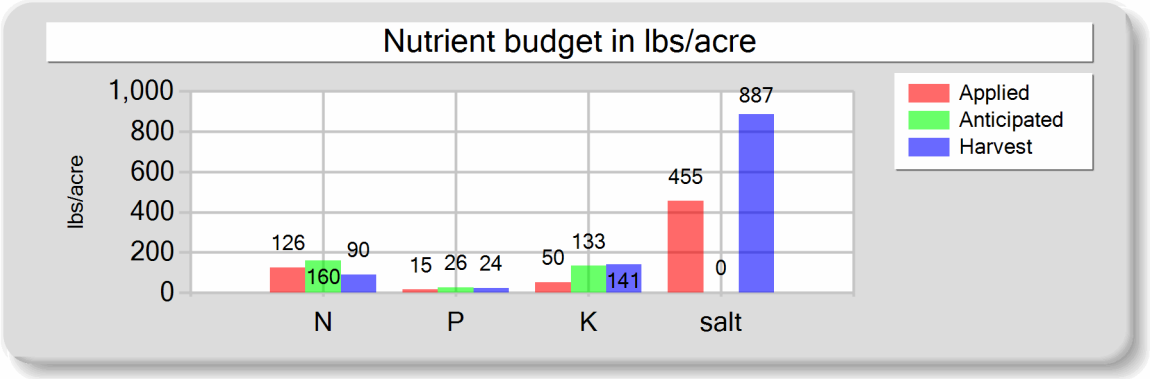
Fresh water applied	
38,724,075.00	*gallons*
1,426.08	*acre-inches*
47.54	*inches/acre*

Process wastewater applied	
1,590,720.00	*gallons*
58.58	*acre-inches*
1.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/10/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	117.16	15.15	50.50	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.59	0.00	0.00	454.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	125.75	15.15	50.50	454.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	89.85	23.85	140.87	887.36
Nutrient balance	35.90	-8.70	-90.37	-432.74
Applied to removed ratio	1.40	0.64	0.36	0.51

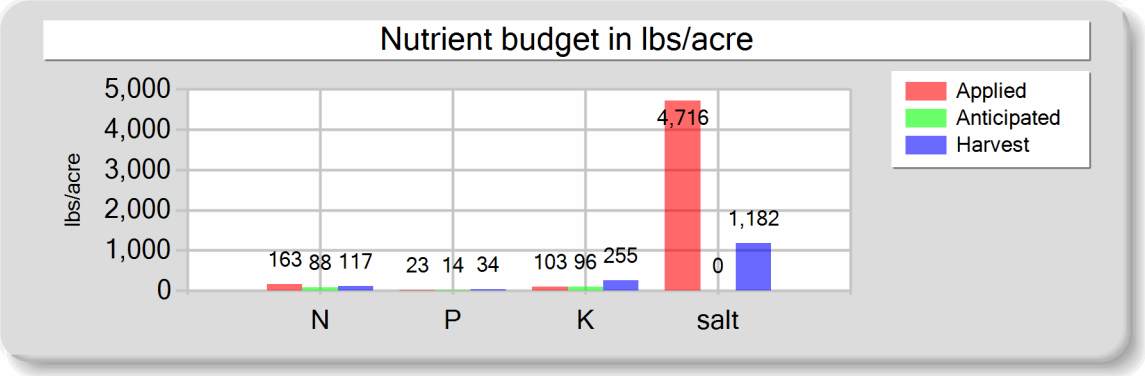
Fresh water applied	
42,493,199.00	*gallons*
1,564.88	*acre-inches*
10.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/15/2023: Sudangrass, silage

Field name: 7 Crop: Sudangrass, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.96	23.33	102.64	2,452.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.92	0.00	0.00	2,263.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.88	23.33	102.64	4,716.10
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	116.52	33.72	254.81	1,182.29
Nutrient balance	46.36	-10.39	-152.17	3,533.81
Applied to removed ratio	1.40	0.69	0.40	3.99

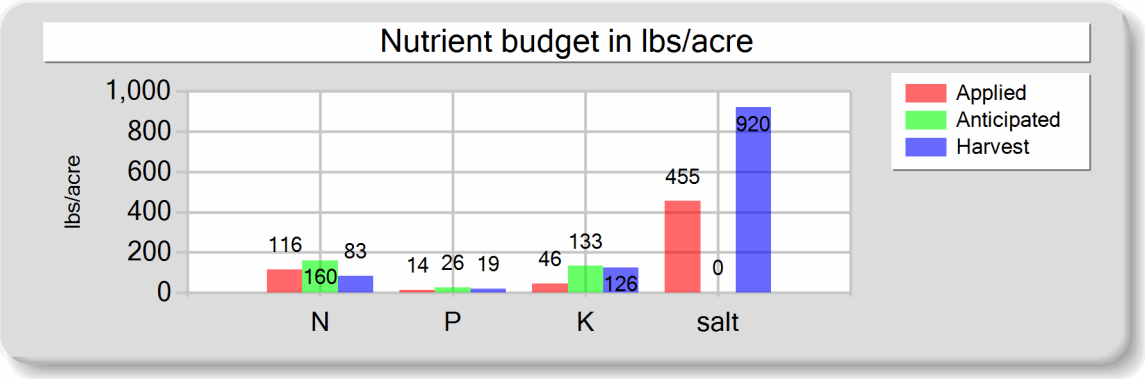
Fresh water applied	
211,564,573.00	*gallons*
7,791.20	*acre-inches*
49.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 10/24/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.36	13.88	46.28	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.59	0.00	0.00	454.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.96	13.88	46.28	454.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	83.25	19.36	125.84	919.60
Nutrient balance	32.71	-5.48	-79.56	-464.98
Applied to removed ratio	1.39	0.72	0.37	0.49

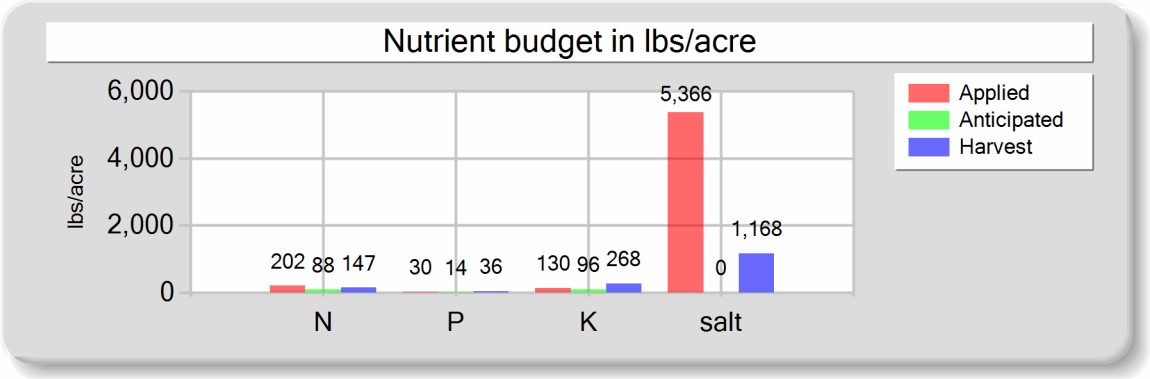
Fresh water applied	
4,903,061.00	*gallons*
180.56	*acre-inches*
10.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/15/2023: Sudangrass, silage

Field name: 8 Crop: Sudangrass, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.15	29.51	129.82	3,102.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.92	0.00	0.00	2,263.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.07	29.51	129.82	5,365.71
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	147.33	35.93	267.71	1,167.86
Nutrient balance	54.74	-6.43	-137.89	4,197.85
Applied to removed ratio	1.37	0.82	0.48	4.59

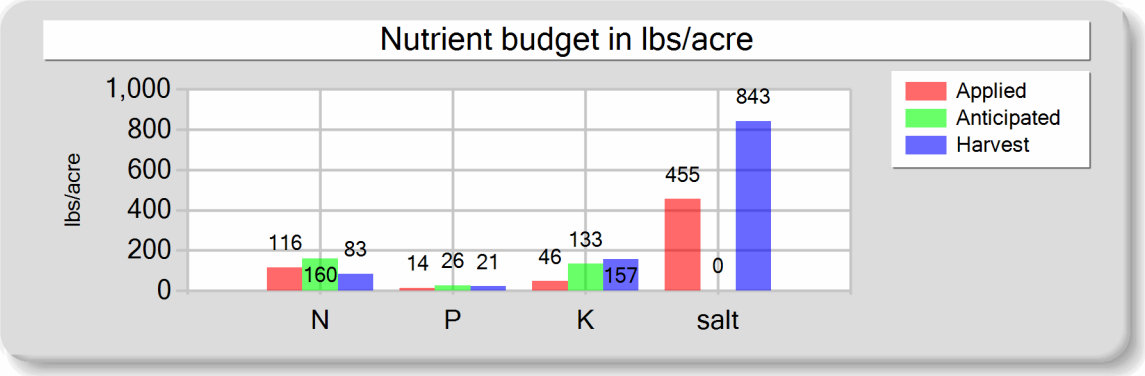
Fresh water applied	
24,411,298.00	*gallons*
898.99	*acre-inches*
49.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/24/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.67	13.92	46.41	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.59	0.00	0.00	454.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.26	13.92	46.41	454.62
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	83.06	21.08	157.45	843.03
Nutrient balance	33.20	-7.15	-111.04	-388.40
Applied to removed ratio	1.40	0.66	0.29	0.54

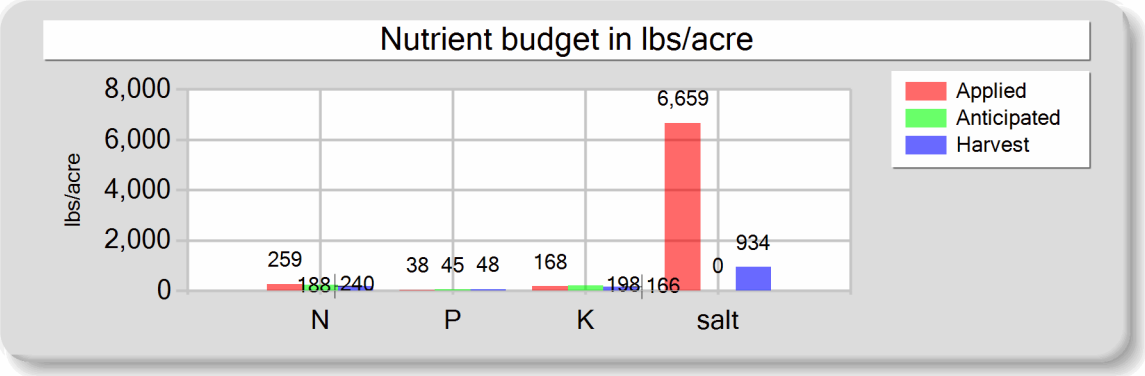
Fresh water applied	
29,418,369.00	*gallons*
1,083.38	*acre-inches*
10.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/28/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.60	38.25	168.29	4,021.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.23	0.00	0.00	2,637.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.83	38.25	168.29	6,658.59
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	187.68	48.05	165.92	933.87
Nutrient balance	71.15	-9.81	2.37	5,724.72
Applied to removed ratio	1.38	0.80	1.01	7.13

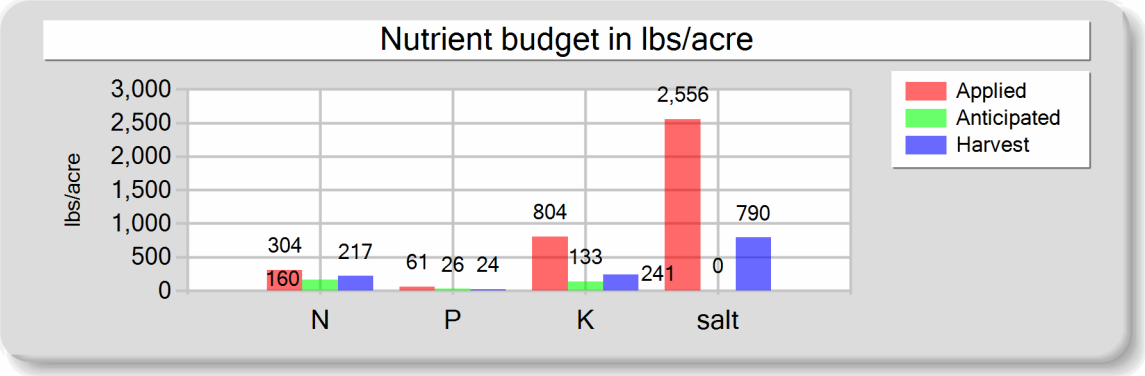
Fresh water applied	
170,649,561.00	*gallons*
6,284.44	*acre-inches*
58.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

E1 - 11/24/2022: Oats, silage-soft dough

Field name: E1 Crop: Oats, silage-soft dough Plant date: 11/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	87.55	31.27	47.19	0.00
Process wastewater	207.79	29.27	757.27	2,147.29
Fresh water	1.43	0.00	0.00	408.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	303.78	60.54	804.45	2,555.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	216.76	24.23	240.55	789.86
Nutrient balance	87.01	36.31	563.91	1,765.84
Applied to removed ratio	1.40	2.50	3.34	3.24

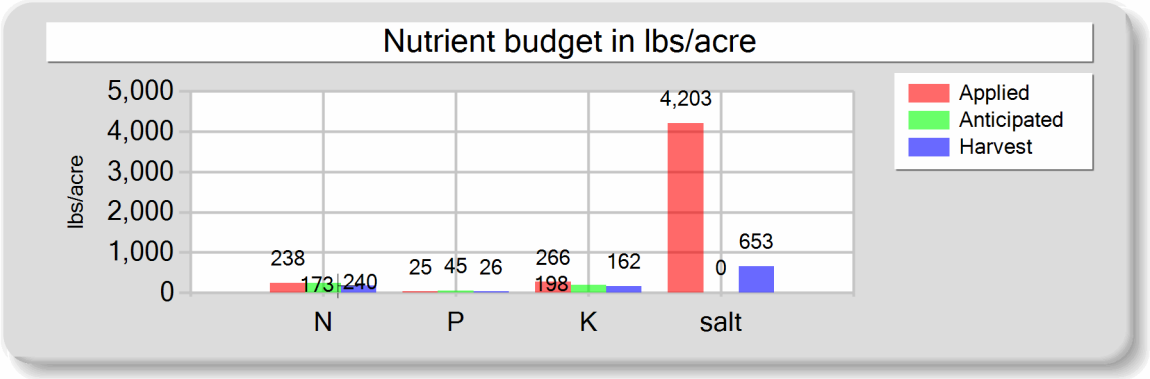
Fresh water applied	
9,298,692.00	*gallons*
342.44	*acre-inches*
9.01	*inches/acre*

Process wastewater applied	
1,665,285.00	*gallons*
61.33	*acre-inches*
1.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

E1 - 06/02/2023: Corn, silage

Field name: E1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.31	24.94	266.04	1,647.09
Fresh water	11.62	0.00	0.00	2,556.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.93	24.94	266.04	4,203.14
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	172.59	26.43	161.70	653.02
Nutrient balance	65.35	-1.50	104.34	3,550.11
Applied to removed ratio	1.38	0.94	1.65	6.44

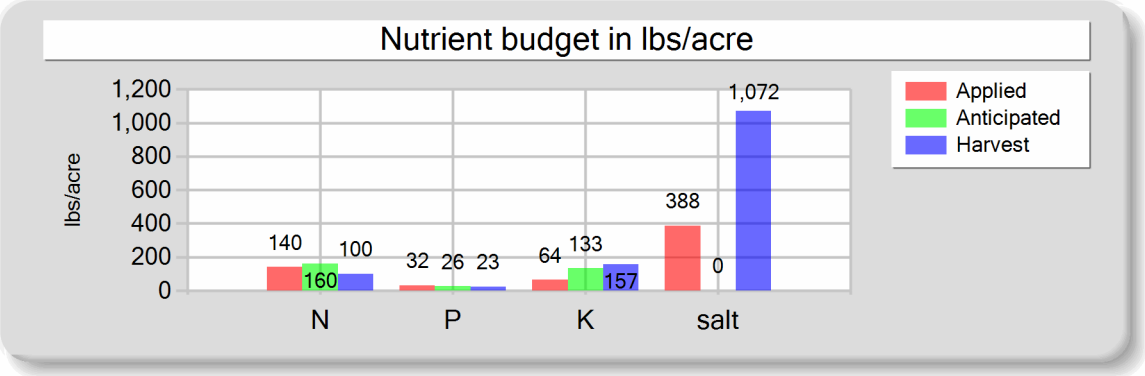
Fresh water applied	
56,135,378.00	*gallons*
2,067.28	*acre-inches*
54.40	*inches/acre*

Process wastewater applied	
1,292,460.00	*gallons*
47.60	*acre-inches*
1.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 1 - 10/25/2022: Oats, silage-soft dough

Field name: Fahey 1 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	131.48	31.99	63.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.36	0.00	0.00	387.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.84	31.99	63.77	387.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	99.85	23.28	156.82	1,072.01
Nutrient balance	39.99	8.71	-93.05	-684.02
Applied to removed ratio	1.40	1.37	0.41	0.36

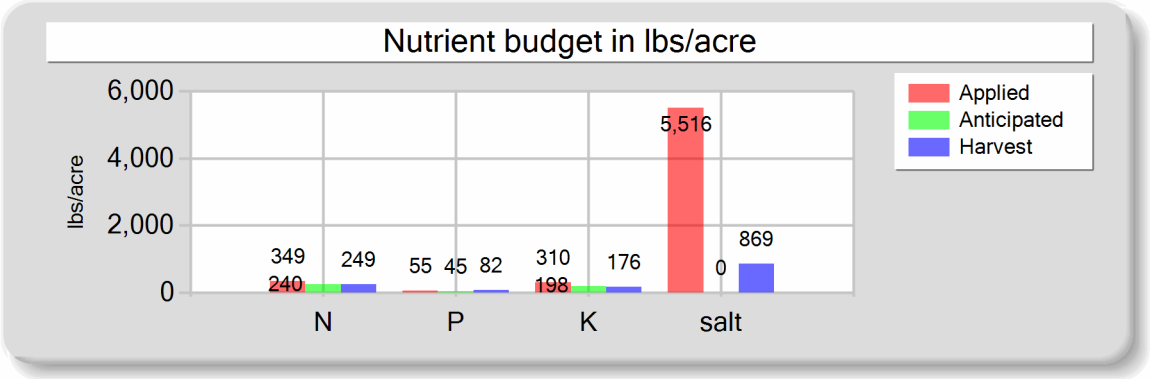
Fresh water applied	
22,549,328.00	*gallons*
830.42	*acre-inches*
8.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 1 - 05/30/2023: Corn, silage

Field name: Fahey 1 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	329.24	54.93	310.42	2,729.12
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	12.41	0.00	0.00	2,786.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.65	54.93	310.42	5,515.76
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.04	82.23	176.21	869.30
Nutrient balance	99.61	-27.30	134.21	4,646.46
Applied to removed ratio	1.40	0.67	1.76	6.35

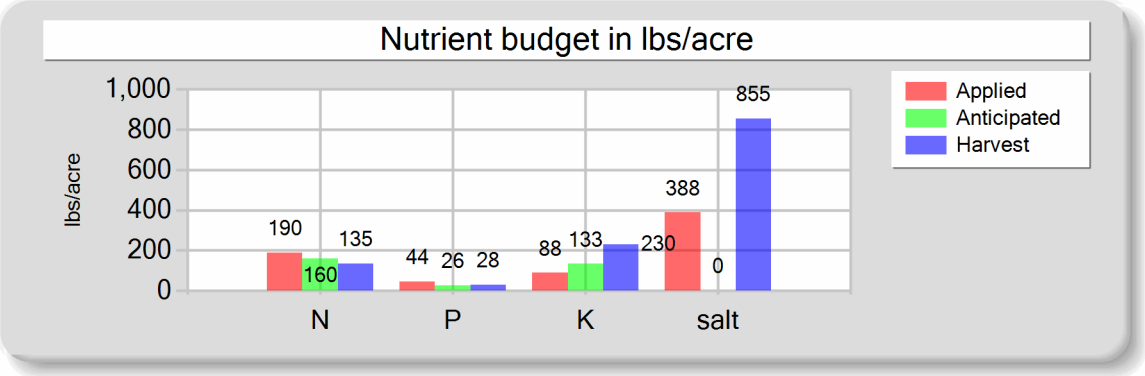
Fresh water applied	
135,588,982.00	*gallons*
4,993.28	*acre-inches*
51.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 2 - 10/26/2022: Oats, silage-soft dough

Field name: Fahey 2 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	181.42	44.13	87.99	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.36	0.00	0.00	387.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.78	44.13	87.99	387.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.47	28.41	229.98	854.64
Nutrient balance	54.30	15.72	-141.99	-466.65
Applied to removed ratio	1.40	1.55	0.38	0.45

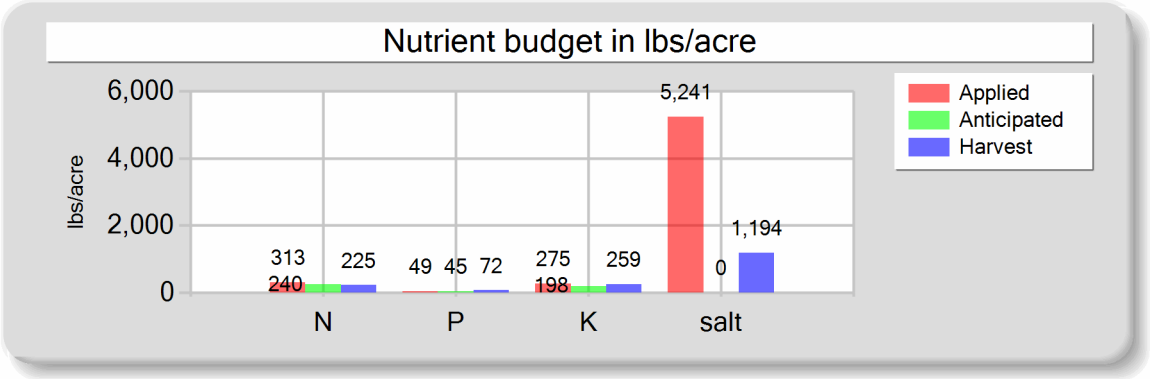
Fresh water applied	
40,681,777.00	*gallons*
1,498.17	*acre-inches*
8.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 2 - 06/03/2023: Corn, silage

Field name: Fahey 2 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	291.25	48.59	274.52	2,416.79
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.04	0.00	0.00	2,824.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.29	48.59	274.52	5,241.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.79	72.02	258.82	1,194.35
Nutrient balance	88.50	-23.43	15.70	4,046.88
Applied to removed ratio	1.39	0.67	1.06	4.39

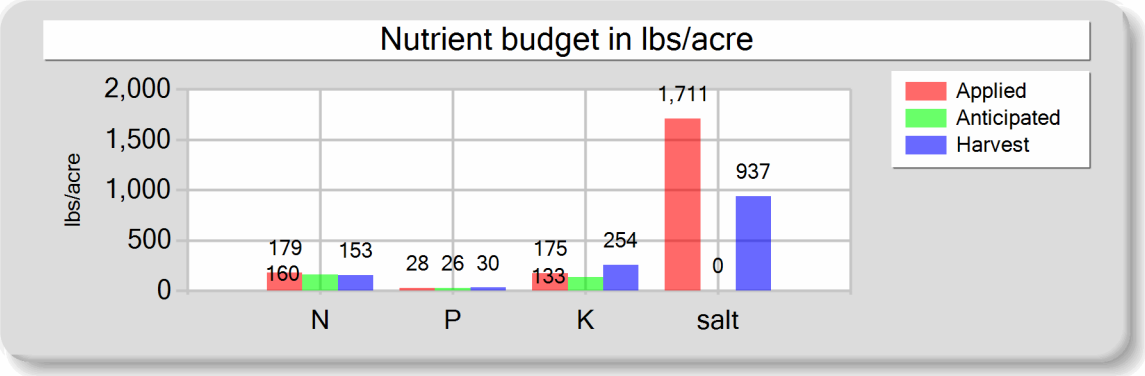
Fresh water applied	
240,877,030.00	*gallons*
8,870.68	*acre-inches*
50.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 3 - 10/26/2022: Oats, silage-soft dough

Field name: Fahey 3 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	170.32	28.34	174.80	1,323.28
Fresh water	1.36	0.00	0.00	387.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.67	28.34	174.80	1,711.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.05	29.81	253.53	937.42
Nutrient balance	25.62	-1.47	-78.73	773.84
Applied to removed ratio	1.17	0.95	0.69	1.83

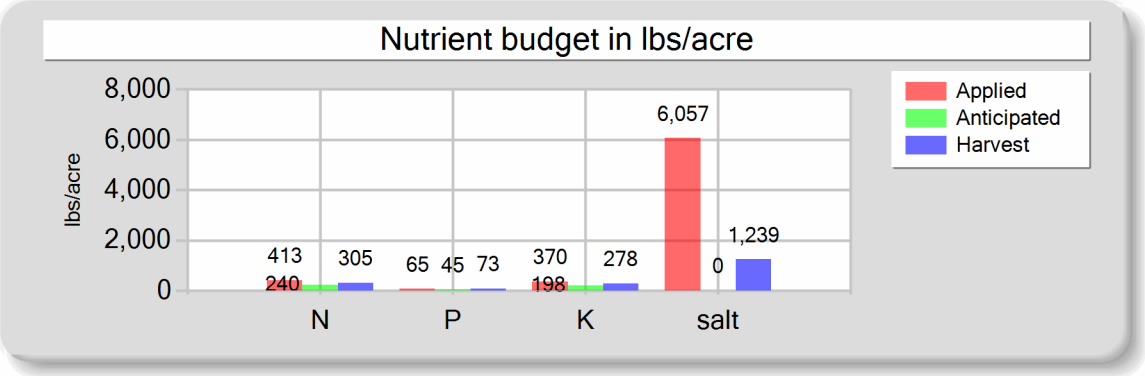
Fresh water applied	
26,036,338.00	*gallons*
958.83	*acre-inches*
8.56	*inches/acre*

Process wastewater applied	
6,900,000.00	*gallons*
254.10	*acre-inches*
2.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fahey 3 - 06/03/2023: Corn, silage

Field name: Fahey 3 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	389.62	65.10	370.07	3,139.17
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.07	0.00	0.00	2,917.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	412.70	65.10	370.07	6,056.96
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	305.16	73.03	277.77	1,238.90
Nutrient balance	107.54	-7.93	92.29	4,818.07
Applied to removed ratio	1.35	0.89	1.33	4.89

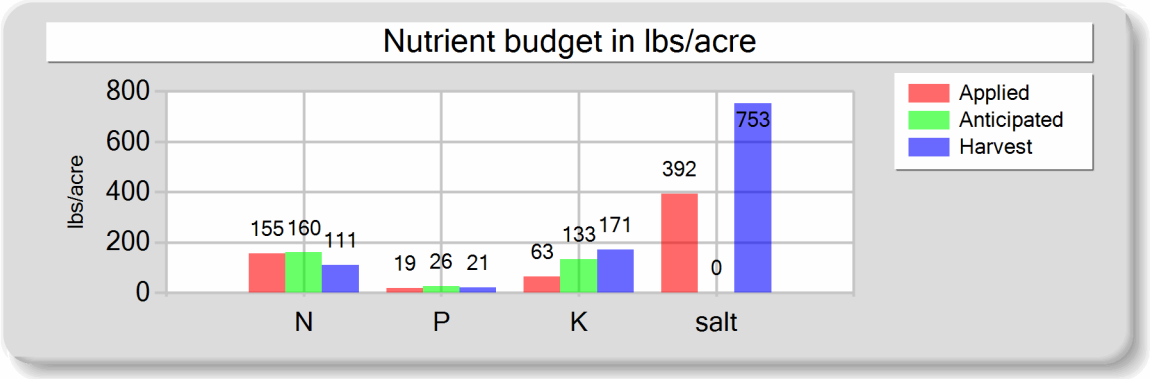
Fresh water applied	
155,601,300.00	*gallons*
5,730.27	*acre-inches*
51.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM1 - 11/22/2022: Oats, silage-soft dough

Field name: JM1 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.74	18.97	63.25	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.37	0.00	0.00	392.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	155.11	18.97	63.25	392.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	110.70	20.62	170.75	752.91
Nutrient balance	44.42	-1.64	-107.50	-360.69
Applied to removed ratio	1.40	0.92	0.37	0.52

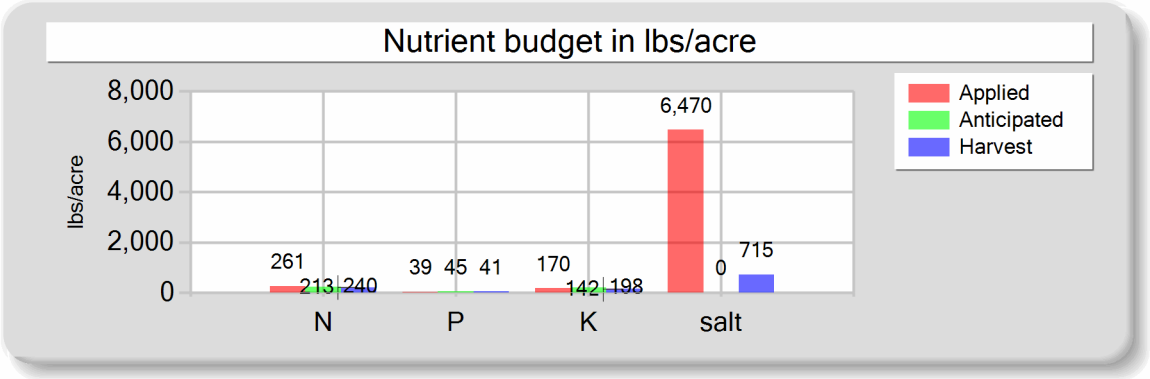
Fresh water applied	
17,155,023.00	*gallons*
631.76	*acre-inches*
8.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM1 - 05/22/2023: Corn, silage

Field name: JM1 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	245.26	38.67	170.13	4,065.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	2,404.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.67	38.67	170.13	6,469.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.64	40.72	142.06	714.82
Nutrient balance	48.04	-2.05	28.07	5,755.13
Applied to removed ratio	1.23	0.95	1.20	9.05

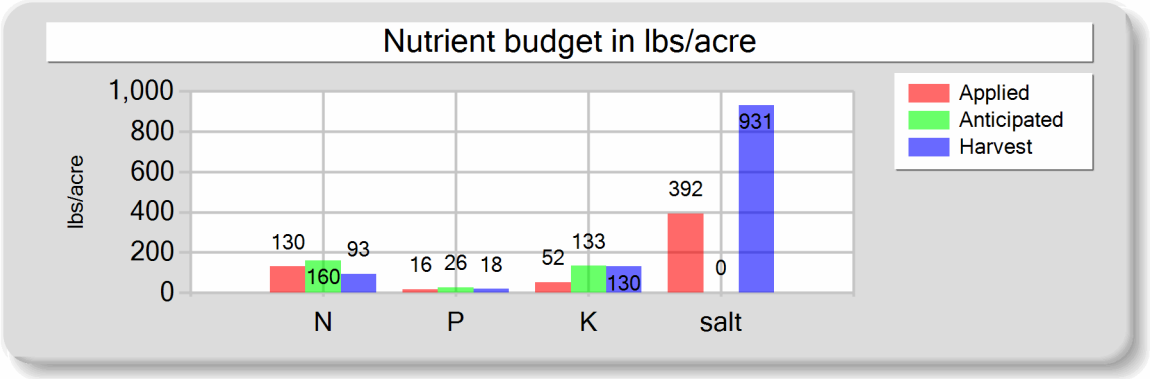
Fresh water applied	
105,167,754.00	*gallons*
3,872.97	*acre-inches*
53.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM2 - 11/22/2022: Oats, silage-soft dough

Field name: JM2 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	121.17	15.67	52.23	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.37	0.00	0.00	392.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	129.54	15.67	52.23	392.22
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	93.14	18.22	130.09	931.37
Nutrient balance	36.40	-2.55	-77.86	-539.15
Applied to removed ratio	1.39	0.86	0.40	0.42

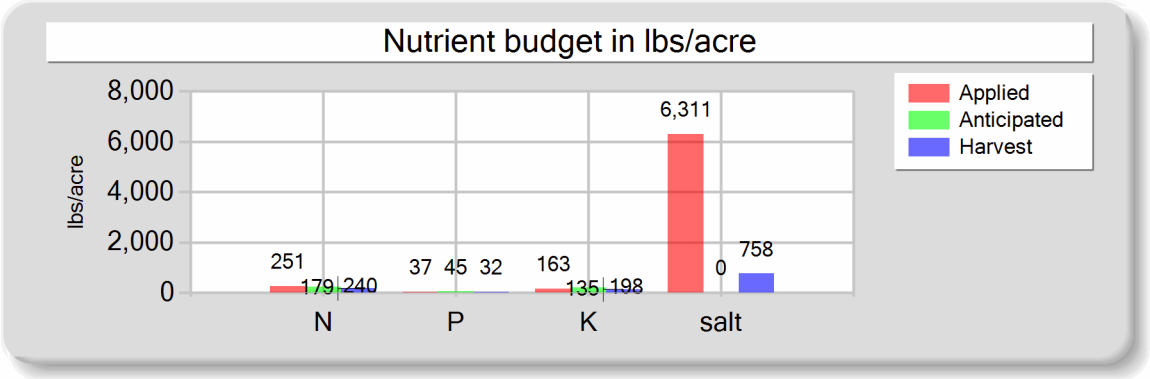
Fresh water applied	
16,920,024.00	*gallons*
623.11	*acre-inches*
8.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM2 - 05/22/2023: Corn, silage

Field name: JM2 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	235.67	37.15	163.48	3,906.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.42	0.00	0.00	2,404.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.08	37.15	163.48	6,310.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	179.41	32.28	135.12	758.20
Nutrient balance	71.67	4.87	28.36	5,552.78
Applied to removed ratio	1.40	1.15	1.21	8.32

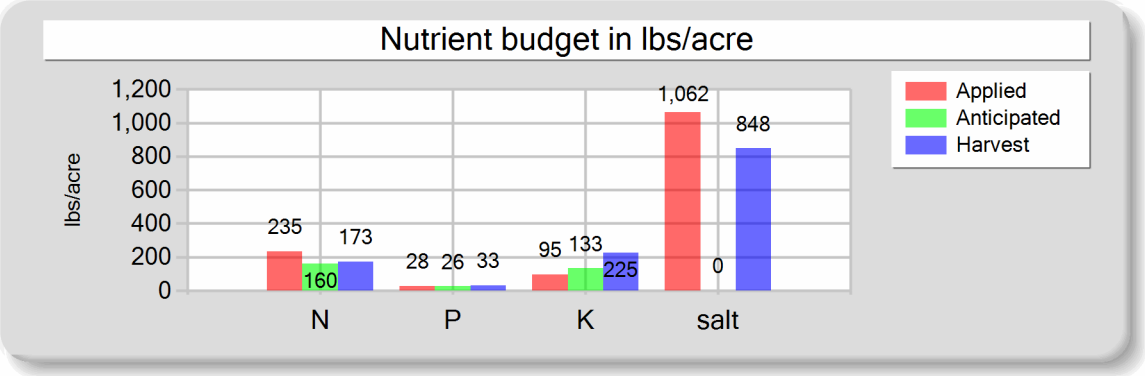
Fresh water applied	
103,727,100.00	*gallons*
3,819.92	*acre-inches*
53.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM3 - 11/22/2022: Oats, silage-soft dough

Field name: JM3 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	219.86	28.43	94.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.30	0.00	0.00	1,062.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.16	28.43	94.77	1,062.15
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.85	32.70	225.31	848.06
Nutrient balance	62.32	-4.27	-130.54	214.10
Applied to removed ratio	1.36	0.87	0.42	1.25

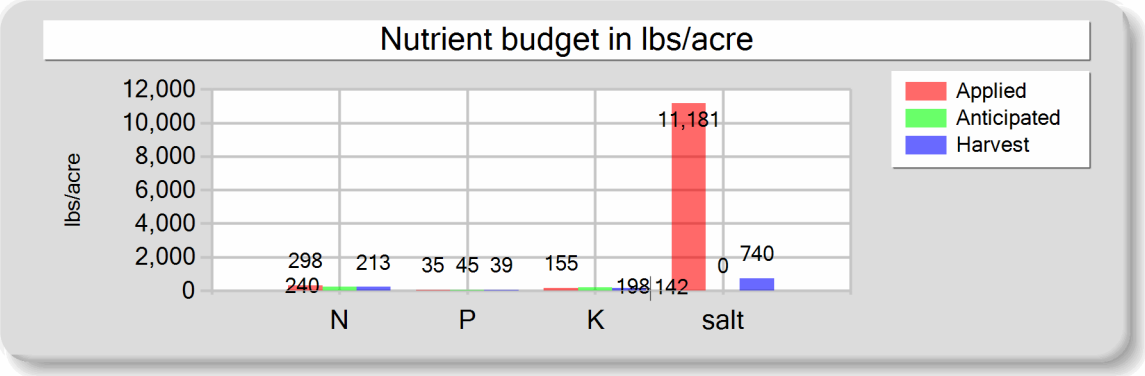
Fresh water applied	
16,274,804.00	*gallons*
599.35	*acre-inches*
10.90	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM3 - 05/21/2023: Corn, silage

Field name: JM3 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	223.44	35.23	155.00	3,703.89
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	67.44	0.00	0.00	7,476.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.88	35.23	155.00	11,180.58
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.68	39.35	142.41	740.17
Nutrient balance	85.20	-4.12	12.59	10,440.42
Applied to removed ratio	1.40	0.90	1.09	15.11

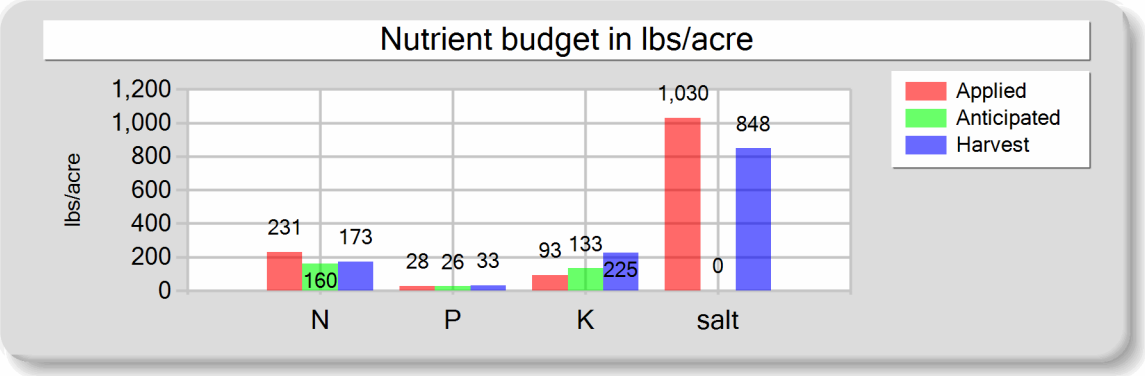
Fresh water applied	
77,841,263.00	*gallons*
2,866.63	*acre-inches*
52.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM4 - 11/22/2022: Oats, silage-soft dough

Field name: JM4 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.18	27.95	93.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.01	0.00	0.00	1,030.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.19	27.95	93.18	1,030.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.84	32.70	225.30	848.02
Nutrient balance	58.35	-4.74	-132.12	182.18
Applied to removed ratio	1.34	0.85	0.41	1.21

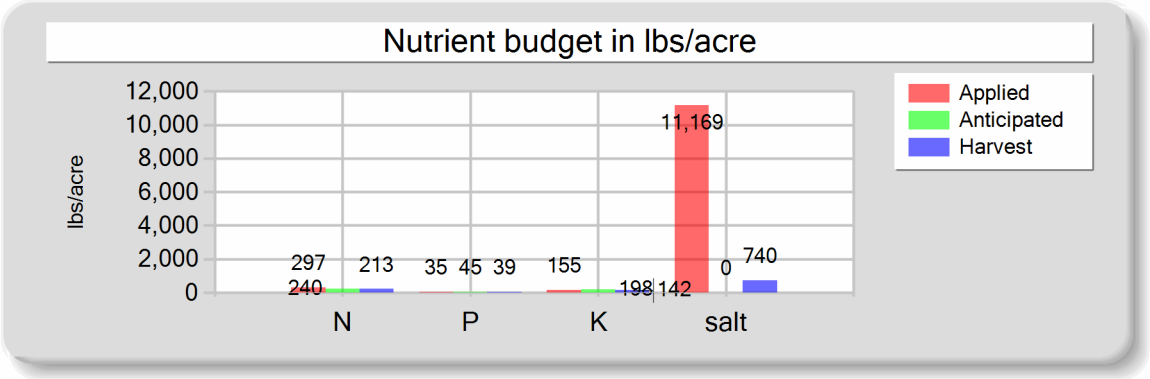
Fresh water applied	
17,140,316.00	*gallons*
631.22	*acre-inches*
10.70	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM4 - 05/21/2023: Corn, silage

Field name: JM4 Crop: Corn, silage Plant date: 05/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	223.09	35.17	154.76	3,698.15
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	67.38	0.00	0.00	7,471.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.48	35.17	154.76	11,169.33
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.69	39.35	142.42	740.21
Nutrient balance	84.79	-4.18	12.34	10,429.12
Applied to removed ratio	1.40	0.89	1.09	15.09

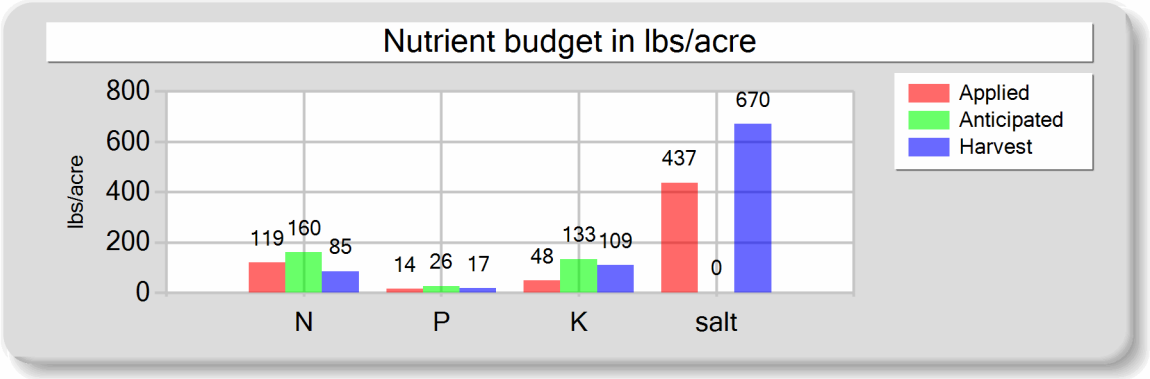
Fresh water applied	
83,447,467.00	*gallons*
3,073.09	*acre-inches*
52.09	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM5 - 11/23/2022: Oats, silage-soft dough

Field name: JM5 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	110.43	14.28	47.60	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.53	0.00	0.00	436.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	118.96	14.28	47.60	436.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	85.12	16.95	108.77	669.93
Nutrient balance	33.84	-2.67	-61.17	-232.95
Applied to removed ratio	1.40	0.84	0.44	0.65

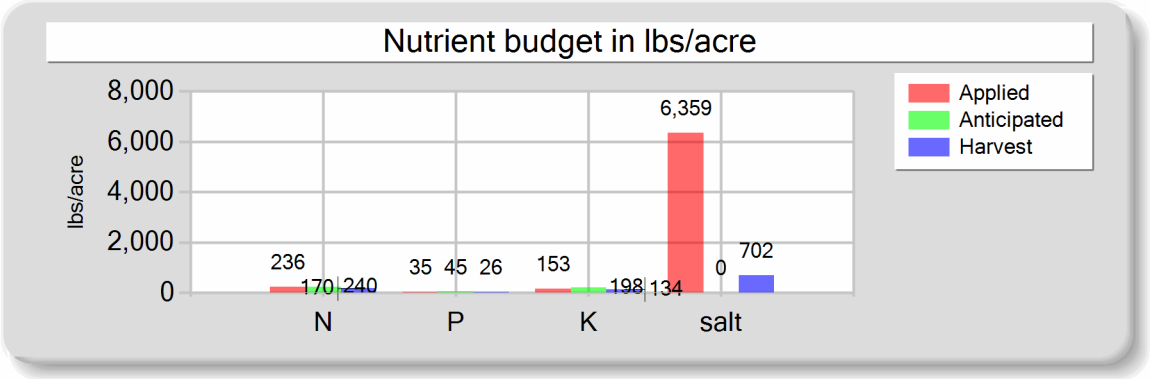
Fresh water applied	
15,447,660.00	*gallons*
568.88	*acre-inches*
9.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

JM5 - 05/24/2023: Corn, silage

Field name: JM5 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	219.92	34.67	152.56	3,645.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.50	0.00	0.00	2,713.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.42	34.67	152.56	6,358.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	170.26	25.82	133.95	702.04
Nutrient balance	66.15	8.85	18.61	5,656.71
Applied to removed ratio	1.39	1.34	1.14	9.06

Fresh water applied	
95,912,370.00	*gallons*
3,532.13	*acre-inches*
59.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Dairy Compost Manure - 2022

Sample and source description: Dairy Compost Manure - 2022

Sample date: 11/22/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	3,000.00	10,000.00							0.00
DL	100.00	100.00	100.00							1.00

Heifer Corral Manure - 2022

Sample and source description: Heifer Corral Manure - 2022

Sample date: 11/22/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	5,500.00	8,300.00							0.00
DL	100.00	100.00	100.00							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	3,500.00	15,400.00	27,700.00	7,900.00	5,000.00	4,800.00	6,000.00		36.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		1.00

Heifer Separator Manure

Sample and source description: Heifer Separator Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,200.00	14,900.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 12/13/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	1,800.00	5,000.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - Dairy - 2022

Sample and source description: Lagoon Q4 - Dairy - 2022

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	586.00	347.00	0.00	1.80	76.10	3,470.00								12,700.00	7,300
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1 - Dairy

Sample and source description: Lagoon Q1 - Dairy

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,710.00	884.00	0.00	1.50	323.00	1,610.00								17,400.00	6,900
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1 - Heifers

Sample and source description: Lagoon Q1 - Heifers

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	473.00	236.00	0.00	1.20	111.00	1,360.00								9,510.00	6,100
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q2 - Heifers

Sample and source description: Lagoon Q2 - Heifers

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,280.00	441.00	0.00	1.00	426.00	2,230.00								14,000.00	3,800
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q2 - Dairy

Sample and source description: Lagoon Q2 - Dairy

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	548.00	352.00	0.00	0.00	84.10	629.00								8,150.00	4,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q3 - Dairy

Sample and source description: Lagoon Q3 - Dairy

Sample date: 07/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	912.00	494.00	0.00	1.10	90.20	1,130.00								14,600.00	6,680
DL	0.01	0.01	0.01	0.01	0.01	0.01								1.00	1

Lagoon Q3 - Heifers

Sample and source description: Lagoon Q3 - Heifers

Sample date: 07/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	219.00	169.00	0.00	0.50	79.40	674.00								5,380.00	3,100
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4 - Dairy

Sample and source description: Lagoon Q4 - Dairy

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	474.00	306.00	0.00	0.10	46.00	448.00								7,110.00	4,220
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4 - Heifers

Sample and source description: Lagoon Q4 - Heifers

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,240.00	500.00	0.00	0.20	152.00	3,050.00								17,200.00	9,850
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

CCID Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70		0.70								328.00	200
DL	0.01		0.01								0.01	1

Domestic Well

DW 1

Sample description: DW 1

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			11.70								2,500.00	
DL			0.01								0.01	

DW 2

Sample description: DW 2

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			1.00								520.00	
DL			0.01								0.01	

Domestic Well

DW 3

Sample description: DW 3

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.90								625.00	
DL			0.01								0.01	

DW 4

Sample description: DW 4

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.50								2,720.00	
DL			0.01								0.01	

DW 6

Sample description: DW 6

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.70								1,040.00	
DL			0.01								0.01	

Domestic Well

DW 7

Sample description: DW 7

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			2.50								2,460.00	
DL			0.01								0.01	

DW 8

Sample description: DW 8

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			1.20								676.00	
DL			0.01								0.01	

DW New

Sample description: DW New

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			10.90	103.00	78.10	303.00	460.00	0.00	440.00	277.00	2,510.00	1,600
DL			0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

Freshwater Pond

Freshwater Pond

Freshwater Pond

Sample description: Freshwater Pond

Sample date: 06/08/2021 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.51										501.00	285
DL	0.01										0.01	1

Irrigation Well

IW Fahey East

Sample description: IW Fahey East

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.60		9.60	75.80	46.70	149.00	438.00	0.00	99.90	145.00	1,410.00	870
DL	0.01		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

IW Fahey West

Sample description: IW Fahey West

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.10		5.10	79.90	38.50	91.00	343.00	0.00	86.30	117.00	1,100.00	670
DL	0.01		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

Irrigation Well

IW Field 1

Sample description: IW Field 1

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.20		10.20	103.00	72.20	339.00	508.00	0.00	392.00	265.00	2,480.00	1,600
DL	0.01		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

IW JM2

Sample description: IW JM2

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.60		6.60	53.50	39.30	138.00	245.00	0.00	152.00	129.00	1,190.00	710
DL	0.01		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

IW JM5

Sample description: IW JM5

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.60		6.60	52.30	38.40	135.00	245.00	0.00	152.00	129.00	1,220.00	740
DL	0.01		0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

D. SOIL ANALYSES

7

7

7 North

Sample and source description: 7 North

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			41.00				
DL			0.01				

7 South

Sample and source description: 7 South

Sample date: 09/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			71.00				
DL			0.01				

8

8

Sample and source description: 8

Sample date: 09/25/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			99.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

1 - 09/20/2019: Alfalfa, hay

1 - 09/20/2019: Alfalfa, hay

Alfalfa - 6th Cut - 1

Sample and source description: Alfalfa - 6th Cut - 1

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	3,300.00	31,200.00		13.90
DL	100.00	100.00	100.00		1.00

Alfalfa - Mid Cut - 1

Sample and source description: Alfalfa - Mid Cut - 1

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,000.00	2,300.00	18,900.00		8.90
DL	100.00	100.00	100.00		1.00

1 - 08/20/2023: Sudangrass, silage

Sudan Silage - 1

Sample and source description: Sudan Silage - 1

Sample date: 10/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	1,700.00	32,100.00		9.80
DL	100.00	100.00	100.00		1.00

10 - 10/29/2022: Oats, silage-soft dough

10 - 10/29/2022: Oats, silage-soft dough

Oat Silage - 10 A

Sample and source description: Oat Silage - 10 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,500.00	2,000.00	20,200.00		7.40
DL	100.00	100.00	100.00		1.00

Oat Silage - 10 B

Sample and source description: Oat Silage - 10 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	3,900.00	36,600.00		12.40
DL	100.00	100.00	100.00		1.00

10 - 06/01/2023: Corn, silage

Corn Silage - 10 A

Sample and source description: Corn Silage - 10 A

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	1,700.00	29,800.00		8.50
DL	100.00	100.00	100.00		1.00

10 - 06/01/2023: Corn, silage

Corn Silage - 10 B

Sample and source description: Corn Silage - 10 B

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	1,400.00	23,600.00		6.80
DL	100.00	100.00	100.00		1.00

11 - 11/01/2022: Oats, silage-soft dough

Oat Silage - 11 A

Sample and source description: Oat Silage - 11 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,400.00	1,800.00	13,700.00		9.30
DL	100.00	100.00	100.00		1.00

Oat Silage - 11 B

Sample and source description: Oat Silage - 11 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,000.00	16,000.00		10.80
DL	100.00	100.00	100.00		1.00

11 - 05/20/2023: Corn, silage

11 - 05/20/2023: Corn, silage

Corn Silage - 11

Sample and source description: Corn Silage - 11

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	2,700.00	11,100.00		5.40
DL	100.00	100.00	100.00		1.00

12N - 11/20/2022: Oats, silage-soft dough

Oat Silage - 12N

Sample and source description: Oat Silage - 12N

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	2,800.00	25,500.00		8.10
DL	100.00	100.00	100.00		1.00

12N - 05/21/2023: Corn, silage

Corn Silage - 12N

Sample and source description: Corn Silage - 12N

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	1,700.00	20,500.00		7.90
DL	100.00	100.00	100.00		1.00

12S - 09/20/2019: Alfalfa, hay

Alfalfa - 6th Cut - 12S

Sample and source description: Alfalfa - 6th Cut - 12S

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,200.00	3,300.00	28,500.00		12.60
DL	100.00	100.00	100.00		1.00

Alfalfa - Mid Cut - 12S

Sample and source description: Alfalfa - Mid Cut - 12S

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 5.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,000.00	2,300.00	15,500.00		8.40
DL	100.00	100.00	100.00		1.00

13 - 10/28/2022: Oats, silage-soft dough

Oat Silage - 13

Sample and source description: Oat Silage - 13

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,600.00	21,000.00		10.40
DL	100.00	100.00	100.00		1.00

13 - 06/01/2023: Corn, silage

13 - 06/01/2023: Corn, silage

Corn Silage - 13

Sample and source description: Corn Silage - 13

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,500.00	14,800.00		6.00
DL	100.00	100.00	100.00		1.00

14 - 11/21/2022: Ryegrass, hay

Ryegrass - 14 A

Sample and source description: Ryegrass - 14 A

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	2,300.00	28,600.00		10.20
DL	100.00	100.00	100.00		1.00

Ryegrass - 14 B

Sample and source description: Ryegrass - 14 B

Sample date: 06/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	2,700.00	30,900.00		11.10
DL	100.00	100.00	100.00		1.00

14 - 07/27/2023: Sudangrass, silage

14 - 07/27/2023: Sudangrass, silage

Sudan Silage - 14 A

Sample and source description: Sudan Silage - 14 A

Sample date: 10/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	2,000.00	31,500.00		12.50
DL	100.00	100.00	100.00		1.00

Sudan Silage - 14 B

Sample and source description: Sudan Silage - 14 B

Sample date: 10/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,000.00	33,100.00		12.10
DL	100.00	100.00	100.00		1.00

2 - 11/01/2022: Oats, silage-soft dough

Oat Silage - 2

Sample and source description: Oat Silage - 2

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,700.00	30,800.00		11.40
DL	100.00	100.00	100.00		1.00

5 - 10/29/2022: Oats, silage-soft dough

5 - 10/29/2022: Oats, silage-soft dough

Oat Silage - 5/6 A

Sample and source description: Oat Silage - 5/6 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,100.00	19,400.00		9.00
DL	100.00	100.00	100.00		1.00

Oat Silage - 5/6 B

Sample and source description: Oat Silage - 5/6 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	2,200.00	23,000.00		6.50
DL	100.00	100.00	100.00		1.00

5 - 05/31/2023: Corn, silage

Corn Silage - 5/6 A

Sample and source description: Corn Silage - 5/6 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	1,800.00	22,300.00		6.90
DL	100.00	100.00	100.00		1.00

5 - 05/31/2023: Corn, silage

Corn Silage - 5/6 B

Sample and source description: Corn Silage - 5/6 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,400.00	19,400.00		7.10
DL	100.00	100.00	100.00		1.00

6 - 10/29/2022: Oats, silage-soft dough

Oat Silage - 5/6 A

Sample and source description: Oat Silage - 5/6 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,100.00	19,400.00		9.00
DL	100.00	100.00	100.00		1.00

Oat Silage - 5/6 B

Sample and source description: Oat Silage - 5/6 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	2,200.00	23,000.00		6.50
DL	100.00	100.00	100.00		1.00

6 - 05/31/2023: Corn, silage

6 - 05/31/2023: Corn, silage

Corn Silage - 5/6 A

Sample and source description: Corn Silage - 5/6 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	1,800.00	22,300.00		6.90
DL	100.00	100.00	100.00		1.00

Corn Silage - 5/6 B

Sample and source description: Corn Silage - 5/6 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,400.00	19,400.00		7.10
DL	100.00	100.00	100.00		1.00

7 - 11/10/2022: Oats, silage-soft dough

Oat Silage - 7A

Sample and source description: Oat Silage - 7A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,400.00	2,300.00	12,300.00		9.10
DL	100.00	100.00	100.00		1.00

7 - 11/10/2022: Oats, silage-soft dough

Oat Silage - 7B

Sample and source description: Oat Silage - 7B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	2,000.00	13,100.00		6.90
DL	100.00	100.00	100.00		1.00

7 - 06/15/2023: Sudangrass, silage

Sudan Silage - 7 A

Sample and source description: Sudan Silage - 7 A

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	4,200.00	29,300.00		14.10
DL	100.00	100.00	100.00		1.00

Sudan Silage - 7 B

Sample and source description: Sudan Silage - 7 B

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	3,700.00	30,400.00		13.60
DL	100.00	100.00	100.00		1.00

8 - 10/24/2022: Oats, silage-soft dough

8 - 10/24/2022: Oats, silage-soft dough

Oat Silage - 8

Sample and source description: Oat Silage - 8

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,600.00	2,000.00	13,000.00		9.50
DL	100.00	100.00	100.00		1.00

8 - 06/15/2023: Sudangrass, silage

Sudan Silage - 8

Sample and source description: Sudan Silage - 8

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	4,000.00	29,800.00		13.00
DL	100.00	100.00	100.00		1.00

9 - 10/24/2022: Oats, silage-soft dough

Oat Silage - 9

Sample and source description: Oat Silage - 9

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,700.00	1,700.00	12,700.00		6.80
DL	100.00	100.00	100.00		1.00

9 - 05/28/2023: Corn, silage

Corn Silage - 9 A

Sample and source description: Corn Silage - 9 A

Sample date: 10/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,500.00	8,600.00		4.40
DL	100.00	100.00	100.00		1.00

Corn Silage - 9 B

Sample and source description: Corn Silage - 9 B

Sample date: 10/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,800.00	9,700.00		5.90
DL	100.00	100.00	100.00		1.00

E1 - 11/24/2022: Oats, silage-soft dough

Oat Silage - E1 A

Sample and source description: Oat Silage - E1 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,800.00	3,100.00	32,800.00		10.70
DL	100.00	100.00	100.00		1.00

E1 - 11/24/2022: Oats, silage-soft dough

Oat Silage - E1 B

Sample and source description: Oat Silage - E1 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,500.00	2,300.00	20,800.00		6.90
DL	100.00	100.00	100.00		1.00

E1 - 06/02/2023: Corn, silage

Corn Silage - E1

Sample and source description: Corn Silage - E1

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	1,700.00	10,400.00		4.20
DL	100.00	100.00	100.00		1.00

Fahey 1 - 10/25/2022: Oats, silage-soft dough

Oat Silage - Fahey 1 A

Sample and source description: Oat Silage - Fahey 1 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	1,800.00	11,500.00		8.60
DL	100.00	100.00	100.00		1.00

Fahey 1 - 10/25/2022: Oats, silage-soft dough

Oat Silage - Fahey 1 B

Sample and source description: Oat Silage - Fahey 1 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,700.00	2,000.00	14,100.00		8.90
DL	100.00	100.00	100.00		1.00

Fahey 1 - 05/30/2023: Corn, silage

Corn Silage - Fahey 1

Sample and source description: Corn Silage - Fahey 1

Sample date: 10/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	3,500.00	7,500.00		3.70
DL	100.00	100.00	100.00		1.00

Fahey 2 - 10/26/2022: Oats, silage-soft dough

Oat Silage - Fahey 2 A

Sample and source description: Oat Silage - Fahey 2 A

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	3,100.00	21,100.00		9.80
DL	100.00	100.00	100.00		1.00

Fahey 2 - 10/26/2022: Oats, silage-soft dough

Oat Silage - Fahey 2 B

Sample and source description: Oat Silage - Fahey 2 B

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,800.00	21,100.00		9.50
DL	100.00	100.00	100.00		1.00

Oat Silage - Fahey 2 C

Sample and source description: Oat Silage - Fahey 2 C

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,900.00	4,500.00	42,000.00		12.00
DL	100.00	100.00	100.00		1.00

Fahey 2 - 06/03/2023: Corn, silage

Corn Silage - Fahey 2A

Sample and source description: Corn Silage - Fahey 2A

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,800.00	11,800.00		4.80
DL	100.00	100.00	100.00		1.00

Fahey 2 - 06/03/2023: Corn, silage

Corn Silage - Fahey 2B

Sample and source description: Corn Silage - Fahey 2B

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	3,100.00	12,400.00		5.90
DL	100.00	100.00	100.00		1.00

Corn Silage - Fahey 2C

Sample and source description: Corn Silage - Fahey 2C

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	3,200.00	8,500.00		4.40
DL	100.00	100.00	100.00		1.00

Fahey 3 - 10/26/2022: Oats, silage-soft dough

Oat Silage - Fahey 3 A

Sample and source description: Oat Silage - Fahey 3 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	3,000.00	25,800.00		10.20
DL	100.00	100.00	100.00		1.00

Fahey 3 - 10/26/2022: Oats, silage-soft dough

Oat Silage - Fahey 3 B

Sample and source description: Oat Silage - Fahey 3 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	2,700.00	23,000.00		8.00
DL	100.00	100.00	100.00		1.00

Oat Silage - Fahey 3 C

Sample and source description: Oat Silage - Fahey 3 C

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	3,200.00	26,900.00		9.80
DL	100.00	100.00	100.00		1.00

Fahey 3 - 06/03/2023: Corn, silage

Corn Silage - Fahey 3A

Sample and source description: Corn Silage - Fahey 3A

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	3,000.00	12,300.00		5.70
DL	100.00	100.00	100.00		1.00

Fahey 3 - 06/03/2023: Corn, silage

Corn Silage - Fahey 3B

Sample and source description: Corn Silage - Fahey 3B

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,600.00	9,000.00		3.80
DL	100.00	100.00	100.00		1.00

JM1 - 11/22/2022: Oats, silage-soft dough

Oat Silage - JM1 A

Sample and source description: Oat Silage - JM1 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,200.00	18,100.00		8.20
DL	100.00	100.00	100.00		1.00

Oat Silage - JM1 B

Sample and source description: Oat Silage - JM1 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,400.00	20,000.00		8.60
DL	100.00	100.00	100.00		1.00

JM1 - 05/22/2023: Corn, silage

JM1 - 05/22/2023: Corn, silage

Corn Silage - JM1 A

Sample and source description: Corn Silage - JM1 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,300.00	7,900.00		3.50
DL	100.00	100.00	100.00		1.00

Corn Silage - JM1 B

Sample and source description: Corn Silage - JM1 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	7,800.00		4.40
DL	100.00	100.00	100.00		1.00

JM2 - 11/22/2022: Oats, silage-soft dough

Oat Silage - JM2 A

Sample and source description: Oat Silage - JM2 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	1,800.00	13,100.00		9.00
DL	100.00	100.00	100.00		1.00

JM2 - 11/22/2022: Oats, silage-soft dough

Oat Silage - JM2 B

Sample and source description: Oat Silage - JM2 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	1,800.00	12,600.00		9.40
DL	100.00	100.00	100.00		1.00

JM2 - 05/22/2023: Corn, silage

Corn Silage - JM2 A

Sample and source description: Corn Silage - JM2 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,400.00	9,700.00		6.10
DL	100.00	100.00	100.00		1.00

Corn Silage - JM2 B

Sample and source description: Corn Silage - JM2 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	1,900.00	8,300.00		4.00
DL	100.00	100.00	100.00		1.00

JM3 - 11/22/2022: Oats, silage-soft dough

JM3 - 11/22/2022: Oats, silage-soft dough

Oat Silage - JM 3 & 4 A

Sample and source description: Oat Silage - JM 3 & 4 A

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	4,800.00	32,700.00		10.90
DL	100.00	100.00	100.00		1.00

Oat Silage - JM 3 & 4 B

Sample and source description: Oat Silage - JM 3 & 4 B

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	4,300.00	30,000.00		12.70
DL	100.00	100.00	100.00		1.00

JM3 - 05/21/2023: Corn, silage

Corn Silage - JM3/JM4 A

Sample and source description: Corn Silage - JM3/JM4 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,100.00	6,700.00		3.70
DL	100.00	100.00	100.00		1.00

JM3 - 05/21/2023: Corn, silage

Corn Silage - JM3/JM4 B

Sample and source description: Corn Silage - JM3/JM4 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,100.00	8,500.00		4.20
DL	100.00	100.00	100.00		1.00

JM4 - 11/22/2022: Oats, silage-soft dough

Oat Silage - JM 3 & 4 A

Sample and source description: Oat Silage - JM 3 & 4 A

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	4,800.00	32,700.00		10.90
DL	100.00	100.00	100.00		1.00

Oat Silage - JM 3 & 4 B

Sample and source description: Oat Silage - JM 3 & 4 B

Sample date: 04/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	4,300.00	30,000.00		12.70
DL	100.00	100.00	100.00		1.00

JM4 - 05/21/2023: Corn, silage

JM4 - 05/21/2023: Corn, silage

Corn Silage - JM3/JM4 B

Sample and source description: Corn Silage - JM3/JM4 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,100.00	6,700.00		3.70
DL	100.00	100.00	100.00		1.00

Corn Silage - JM3/JM4 B

Sample and source description: Corn Silage - JM3/JM4 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,100.00	8,500.00		4.20
DL	100.00	100.00	100.00		1.00

JM5 - 11/23/2022: Oats, silage-soft dough

Oat Silage - JM5 A

Sample and source description: Oat Silage - JM5 A

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,300.00	13,200.00		8.90
DL	100.00	100.00	100.00		1.00

JM5 - 11/23/2022: Oats, silage-soft dough

Oat Silage - JM5 B

Sample and source description: Oat Silage - JM5 B

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,800.00	2,000.00	14,400.00		8.10
DL	100.00	100.00	100.00		1.00

JM5 - 05/24/2023: Corn, silage

Corn Silage - JM5 A

Sample and source description: Corn Silage - JM5 A

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,400.00	1,400.00	8,800.00		4.10
DL	100.00	100.00	100.00		1.00

Corn Silage - JM5 B

Sample and source description: Corn Silage - JM5 B

Sample date: 09/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	1,800.00	7,800.00		4.60
DL	100.00	100.00	100.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

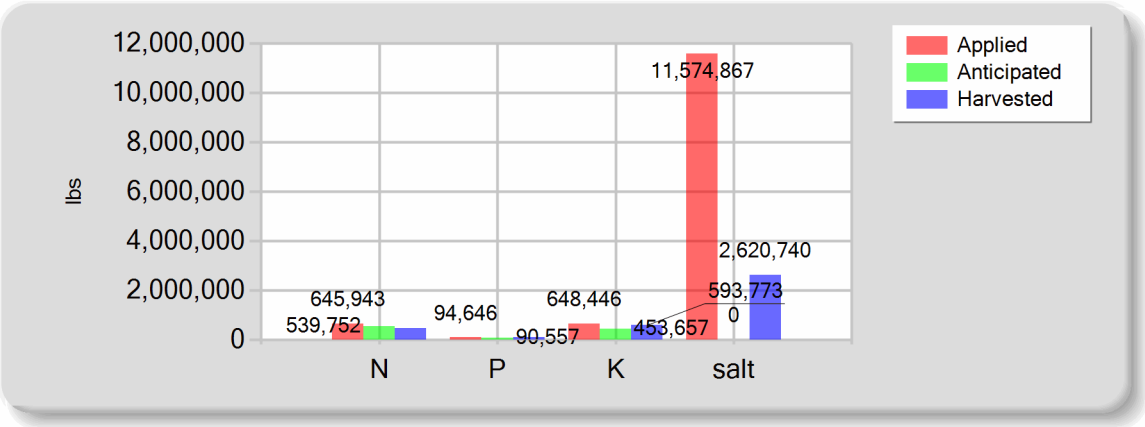
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

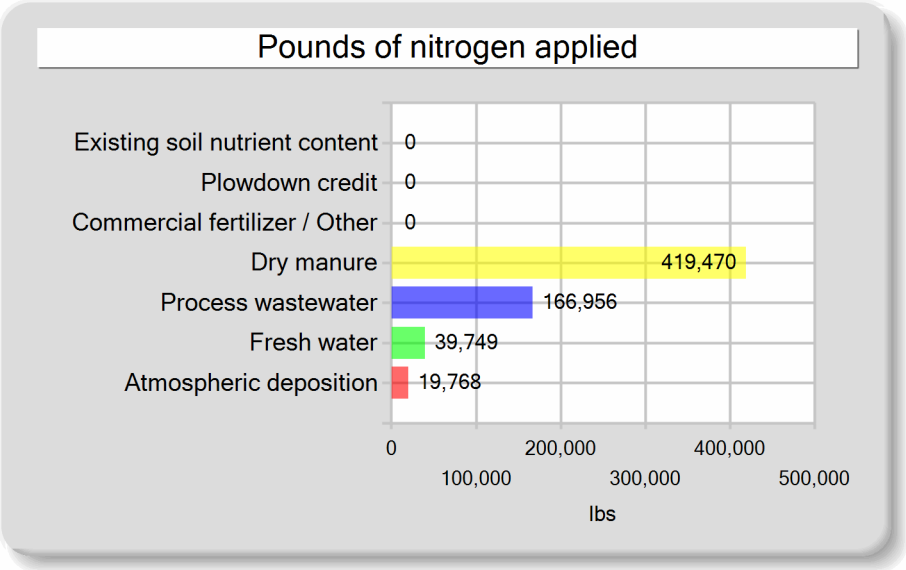
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

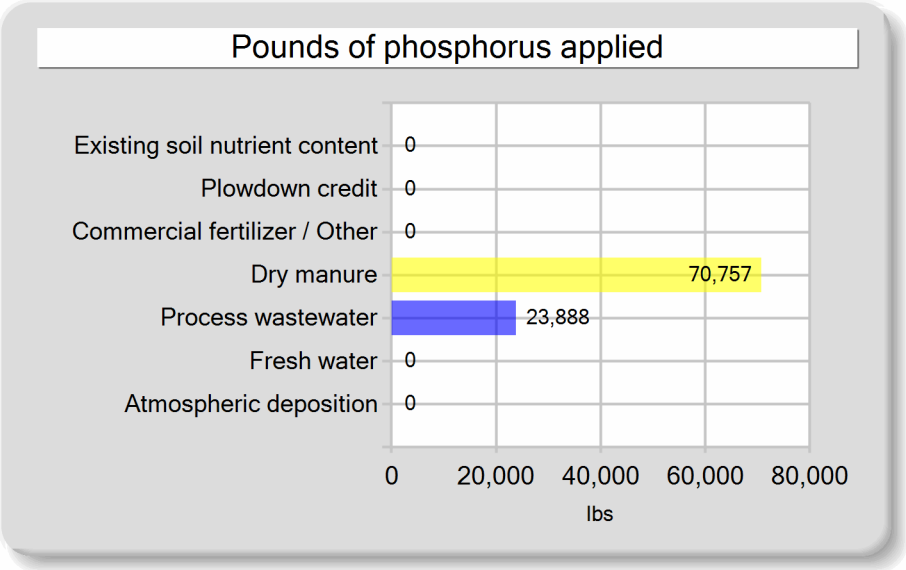
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	419,469.78	70,757.15	295,680.58	3,345,662.43
Process wastewater	166,956.18	23,888.41	352,765.89	1,400,154.72
Fresh water	39,748.92	0.00	0.00	6,829,049.41
Atmospheric deposition	19,768.00	0.00	0.00	0.00
Total nutrients applied	645,942.88	94,645.56	648,446.47	11,574,866.56
Anticipated crop nutrient removal	539,752.00	90,557.20	453,656.80	0.00
Actual crop nutrient removal	475,230.05	97,223.49	593,772.52	2,620,740.17
Nutrient balance	170,712.84	-2,577.93	54,673.95	8,954,126.39
Applied to removed ratio	1.36	0.97	1.09	4.42

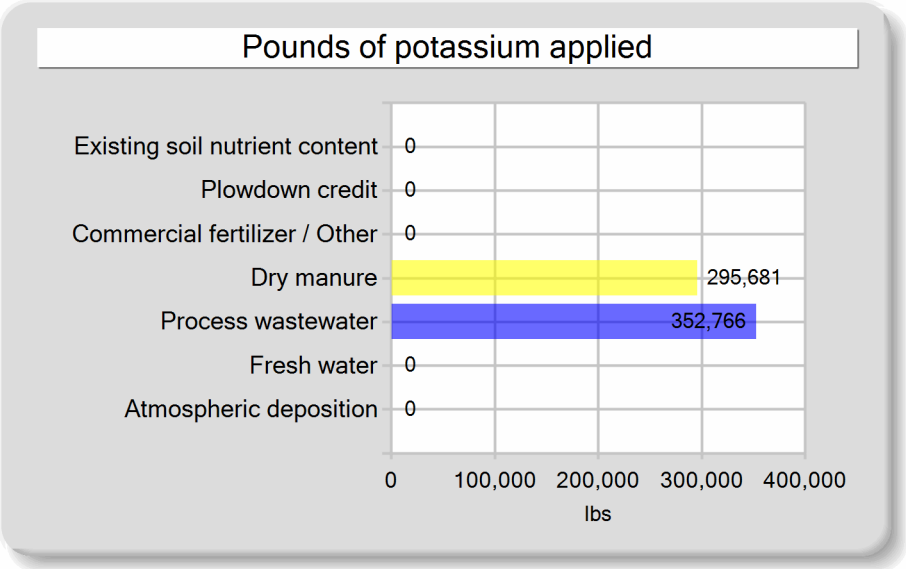
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

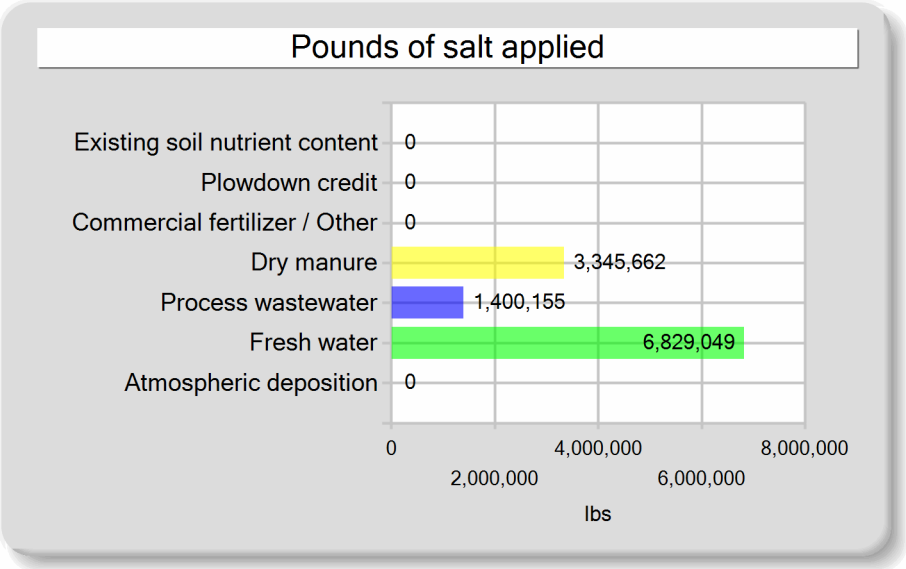


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 2 taken out of production after oat harvest

Over application occurred on Field 14 during last few checks of irrigation event, see PROSE report previously submitted. Wastewater reported as discharged on PROSE report was to reflect only the area of discharge, not calculated wastewater applied over entire field

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Manuela Godinho	David Godinho
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuela Godinho	David Godinho
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	Jun 26, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Godinho Dairy
12710 S Wilson Road
Los Banos CA 93635
00-0021352 08

E-mail: godinhodairy209@gmail.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L1002
Sampled By: M Pedroso
Requested By: David Godinho
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TDS mg/L	CO_3 as $CaCO_3$ mg/L	HCO_3 as $CaCO_3$ mg/L	Cl mg/L	SO_4 mg/L	Ca mg/L	Mg mg/L	Na mg/L	pH at 25°C unit
1	IW Fahey West	12/14/23	8:00	1100	1.10	5.1	ND		670	ND	343	117	86.3	79.9	38.5	91	7.9
2	IW Fahey East	12/14/23	8:05	1410	1.41	9.6	ND		870	ND	438	145	99.9	75.8	46.7	149	7.9
3	IW Field 1	12/14/23	8:30	2480	2.48	10.2	ND		1600	ND	508	265	392	103	72.2	339	7.7
4	IW JM 5	12/14/23	8:40	1220	1.22	6.6	ND		740	ND	245	129	152	52.3	38.4	135	7.9
5	DW - New	12/14/23	8:45	2510	2.51	10.9	ND		1600	ND	460	277	440	103	78.1	303	7.9
6	IW - JM 2	12/14/23	8:55	1190	1.19	6.6	ND		710	ND	245	129	152	53.5	39.3	138	7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

12/14/23 14:20 23L1002

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21352 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Godinho Dairy
Address: 12710 S Wilson Road
City, State, Zip: Los Banos, CA 93635
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: David Godinho - (209) 587-1541

Ranch _____

Date sampled: 12/14/23

Sampled by: MPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 6 No. Bottles: 6

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

( ) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

(X) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES	Sampled From:	Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. 1W Fahey West		12/14/23	800A	0	5.1
2. 1W Fahey East		12/14/23	805A	0	5.4
3. 1W Field 1		12/14/23	830A	0	5.7
4. 1W Jm5		12/14/23	840A	0	5.8
5. DW-new		12/14/23	845A	0	5.2
6. 1W -Jm2		12/14/23	855A	0	5.3
7.					
8.					
9.					
10.					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPed	CASINC	12/14/23 855A	12/14/23 12pm
Second	CSM	DLI	12/14/23 12:00	12/14/23 2:15pm
Third				
Fourth	[illegible]	DVE	12/14 1420	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Lab No.: 23L1008
Sampled By: M Dekeyrel
Requested By: David Godinho
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

Godinho Dairy
12710 S Wilson Road
Los Banos CA 93635
00-0021352 08

E-mail: godinhodairy209@gmail.com
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW #1	12/13/23	15:09	2500	2.50	11.7	ND		7.5
2	DW #2	12/13/23	15:07	520	0.52	1.0	ND		8.1
3	DW #3	12/13/23	14:50	625	0.62	0.9	ND		7.9
4	DW #4	12/13/23	14:39	2720	2.72	2.5	ND		7.6
5	DW #6	12/13/23	15:13	1040	1.04	2.7	ND		7.6
6	DW #7	12/13/23	14:41	2460	2.46	2.5	ND		8.4
7	DW #8	12/13/23	14:43	676	0.68	1.2	ND		7.9

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

12/14/23 14:20 23L1008

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WATER WORK REQUEST

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Address: 12710 S Wilson Road
City, State, Zip: Los Banos, CA 93635
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: David Godinho - (209) 587-1541

Ranch _____

Date sampled: 12/13/23

Sampled by: M. Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 7 No. Bottles: 7

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**)
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW #1	Sampled From:	12/13/23	3:09pm	Ø	5.7
2. DW #2	Sampled From:	12/13/23	3:07pm	Ø	7.1
3. DW #3	Sampled From:	12/13/23	2:50pm	Ø	7.1
4. DW #4	Sampled From:	12/13/23	2:39pm	Ø	4.9
5. DW #6	Sampled From:	12/13/23	3:13pm	Ø	0.6
6. DW #7	Sampled From:	12/13/23	2:41pm	Ø	6.1
7. DW #8	Sampled From:	12/13/23	2:43pm	Ø	5.4
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

- IR Thermometer SN: 221314357
  Correction Factor: 0°C
- Calibration Due: 03/06/2024
  Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M. Dekeyrel	CASINC	12/13/23 @ 3:14pm	12/14/23 12p
Second	JLF	DLV	12-14-23 - 12:00	2-14-23 2:15
Third				
Fourth	[illegible]	DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

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Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020